



26MWP SOLAR PHOTOVOLTAIC POWER PLANT IN BAVET CITY, CAMBODIA



Document Prepared by CTI Certification Co., Ltd

Project Title	26MWp Solar Photovoltaic Power Plant in Bavet City, Cambodia
Report Title	Joint Validation and 1 st periodical VCS Verification report for 26MWp Solar Photovoltaic Power Plant in Bavet City, Cambodia
Version	2.0
Report ID	2022-11001
Verification Period	27-January-2022 to 31-December-2022
Client	Climate Bridge (Shanghai) Ltd.
Pages	76
Date of Issue	18-December-2023
Prepared By	CTI Certification Co., Ltd
Contact	CTI Certification Co., Ltd F8-A CTI Building, No.4 LiuXianSan Road, Xin'an Street Bao'an District, Shen Zhen 518101, P. R. China Phone: +86-10-65580012

	https://www.cti-cert.org/ Contact: Ms. Lin Shunrong Email: linshunrong@cti-cert.com
Approved By	Zhou Lu 
Work Carried Out By	Li Shunrong (TL& verifier) Emanuel Rogier van Mansvelt (Local Expert) Lin Wu (TR)

Summary:

- *A description of the project*

The project activity developed in Monourom village, Bavet City, Svay Rieng Province, the Kingdom of Cambodia, is to install a solar power project with total installed capacity of 26MWp. The utilization of solar energy at the project site is for generation of power, the average annual power delivered to the grid by the project is expected to be 37,085 MWh. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants in Electricite Du Cambodge (EDC) Grid. The project is expected to achieve 22,790 tCO_{2e} of emission reduction annually. The project started operation on 27-January-2022 with the VCS Ref. No. 4036.

- *A description of the validation and verification*

The approved CDM methodology ACM0002 "Grid-connected electricity generation from renewable sources" (version 21.0) is applied to quantify the GHG emission reductions achieved in this project. The calculation of the GHG emission reductions is carried out in a transparent and conservative manner. This project is being developed in conjunction with the validation and 1st periodic of verification.

- *The purpose and scope of validation and verification*

The validation objective is an independent assessment by a Third Party of a proposed project activity against all defined criteria set for the registration under the VCS. In order to confirm that the project activity, as documented, is sound reasonable and meets the identified criteria, the validation involves the assessment of: project conformance to VCS standards/programs, project conformance to the applied methodology, including the procedure for the demonstration of additionality specified in the methodology; and likelihood that methods and procedures set out in the project description will generate verifiable GHG data and information when implemented. Validation is a requirement and is seen as necessary to provide assurance to stakeholders of the quality of project and its intended generation of VCUs. Validation is part of the VCS project cycle and will finally result in a conclusion by the executing VVB whether a project activity is valid to be submitted for registration to VCS registry.

The objective of the verification is to have an independent review ex post determination by a VVB (Validation and Verification Body) of the monitored GHG emission reductions that have occurred as a result of the implementation of the project activity during a defined monitoring period.

- *The method and criteria used for validation and verification*

Validation and Verification is conducted using CTI Certification Co., Ltd (CTI) procedures in line with the requirements specified in the latest version of the VCS Validation and Verification Manual and applying auditing techniques. The validation/verification team assessed the project activity's compliance against the VCS Standard Version 4.4, the selected CDM methodology and the joint project description and monitoring report. The project is eligible under Project Scope 1. The validation/verification criteria followed the guidance documents provided by VCS included the following:

VCS Standard Version 4.4, VCS Program Guide Version 4.3 and the applied CDM methodology ACM0002 "Grid-connected electricity generation from renewable sources" (version 21.0).

- *The number of findings raised during validation and verification*

In the course of the validation and verification, 12 Corrective Action Requests (CARs), 11 Clarification Requests (CLs) were raised and successfully closed. The assessment is included in the report.

- *Any uncertainties associated with the validation and verification*

There are no restrictions of uncertainty for both validation and verification.

- *Summary of the validation and verification conclusions*

Climate Bridge (Shanghai) Ltd. has commissioned the CTI Certification Co., Ltd to carry out the Verified Carbon Standard (VCS) validation joint with 1st periodical verification of the project, “26MWp Solar Photovoltaic Power Plant in Bavet City, Cambodia” (VCS ID 4036) with regard to the relevant requirements of VCS Standard Version 4.4. CTI confirms all validation and verification activities including objectives, scope and criteria, level of assurance, project description, monitoring and monitoring report adhere to VCS Standard Version 4.4 and all associated updated as documented in this report, are complete.

CTI concludes that the project activity “26MWp Solar Photovoltaic Power Plant in Bavet City, Cambodia” in Cambodia, as described in the final version of Joint-Project-Description-Monitoring-Report, meets all relevant requirements for VCS validation and verification activity and correctly applied the methodology ACM0002 Version 21.0. Hence CTI is able to certify that the GHG emission reductions from the project during the monitoring period from 27-January-2022 to 31-December-2022 were generated as 20,393 tCO₂e which can be claimed as VCU.

CONTENTS

1	INTRODUCTION.....	6
1.1	Objective	6
1.2	Scope and Criteria	6
1.3	Reasonableness of Assumptions and Level of Assurance	7
1.4	Summary Description of the Project.....	7
2	VALIDATION AND VERIFICATION PROCESS	9
2.1	Method and Criteria.....	9
2.2	Document Review.....	9
2.3	Interviews	10
2.4	Site Visits	11
2.5	Resolution of Findings	12
2.5.1	Forward Action Requests.....	12
3	VALIDATION FINDINGS	13
3.1	Project Details	13
3.2	Participation under Other GHG Programs	20
3.3	Safeguards	20
3.4	Application of Methodology	24
3.5	Non-Permanence Risk Analysis.....	49
4	VERIFICATION FINDINGS.....	50
4.1	Accuracy of GHG Emission Reduction and Removal Calculations	50
4.2	Quality of Evidence to Determine GHG Emission Reductions and Removals	53
5	VALIDATION AND VERIFICATION OPINION	54
	APPENDIX 1: ABBREVIATIONS.....	56
	APPENDIX 2: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS	57
	APPENDIX 3: DOCUMENTS REVIEWED OR REFERENCED	57
	APPENDIX 4: CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS AND FORWARD ACTION REQUESTS.....	63

1 INTRODUCTION

1.1 Objective

Climate Bridge (Shanghai) Ltd. has commissioned the CTI Certification Co., Ltd to carry out the Verified Carbon Standard (VCS) validation joint with 1st periodical verification of the project for the monitoring period from 27-January-2022 to 31-December-2022 (both days included).

The objective of validation is an independent assessment by a Third Party of a proposed project activity against all defined criteria set for the registration under the VCS. In order to confirm that the project activity, as documented, is sound reasonable and meets the identified criteria, the validation involves the assessment of: project conformance to VCS standards/programs, project conformance to the applied methodology, including the procedure for the demonstration of additionality specified in the methodology; and likelihood that methods and procedures set out in the project description will generate verifiable GHG data and information when implemented. Validation is a requirement and is seen as necessary to provide assurance to stakeholders of the quality of project and its intended generation of VCU. Validation is part of the VCS project cycle and will finally result in a conclusion by the executing VVB whether a project activity is valid to be submitted for registration to VCS registry.

The objective of verification is to have an independent review ex-post determination by a VVB of the monitored GHG emission reductions that have occurred as a result of the project activity implemented during a defined monitoring period. The evaluation is done against the requirements of the VCS Standard Version 4.4 based on the monitoring report. In order to confirm that the GHG emission reductions sound reasonable and meet the identified criteria, the verification involves the assessment of monitoring plan conformance to VCS rules and applied methodology. Verification is a requirement and is seen as necessary to provide assurance on the generation of VCUs.

1.2 Scope and Criteria

The validation scope is defined as an independent and objective review of the project description (PD) to validate that (a) the project design is actual, (b) the baseline scenario is correctly defined as per the applied methodology and relate tools, (c) the project is additional, (d) the monitoring plan can be implemented and is transparent and adequate and (e) all data and information used for ex-ante calculation of emission reductions is of projected and/or hypothetical nature. The Joint-PD-MR is reviewed against the criteria stated in VCS Standard Version 4.4/41/ and the approved baseline and monitoring methodology ACM0002 Version 21.0/31/. The validation was based on the requirements VCS Validation and Verification Manual and VCS Standard Version 4.4 applying auditing techniques. The validation is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

The scope of the verification is to verify that: (a) the actual monitoring system and procedures are in compliance with the monitoring systems and procedures described in the monitoring plan; (b) the GHG emission reductions data and express a conclusion with a reasonable level of assurance about whether the reported GHG emission reductions data are free from material misstatement; (c) the reported GHG emissions data is sufficiently supported by evidence. Verification shall ensure that the reported GHG emission reductions are complete and accurate in accordance with the applicable VCS criteria in order to be certified. Verification is conducted using CTI procedures in line with the requirements specified in the VCS Program and ISO14064-3 requirements and applying auditing techniques. The VVB assessed and determined that the implementation and operation of the project activity, and steps to report GHG emission reductions comply with the VCS rules. The verification involved a document review of relevant documentation and interview with the project participants. Verification is not meant to provide any consultancy towards the project participants; however, stated request for clarifications and/or corrective actions may have provided input for improvement of the monitoring.

1.3 Reasonableness of Assumptions and Level of Assurance

Based on the validated assumptions, limitations, and methods used during the validation (see section 3.4.6 for ex ante ER calculation result), the project is expected to generate average GHG emission removal of 22,790 tCO_{2e} per year, and it is confirmed that outcome of future activities will be achieved as the assumptions and limitations are reasonable.

All GHG sinks, sources and GHG emissions equal to or greater than 5% of the total GHG assertions are considered. The project is expected to generate average GHG emission removal of 22,790 tCO_{2e} per year, and therefore can be considered as a project per the VCS rules, in regards to its scale, subject to a 5% materiality threshold.

The assessment was conducted to provide a reasonable level of assurance of compliance against the defined audit criteria and materiality threshold within the audit scope. Based on the audit findings the validation and verification statement reasonably assure that the GHG assertion is materially correct and is a fair representation of the GHG data and information.

1.4 Summary Description of the Project

The project is implemented by Green Sustainable Ventures (Cambodia) Co., Ltd., the main purpose of this project activity is to generate clean form of electricity through renewable solar energy. The installed capacity of the project activity is 26MWp, consisting of 57,639 pieces of the solar PV cell modules with unit capacity of 455Wp. The utilization of solar energy from project site is for generation of power, the average annual power delivered to the grid by the project is expected to be 37,085 MWh. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants in Electricite Du Cambodge Grid (EDC). It's estimated that the project activity could achieve GHG emission reductions of 22,790 tCO_{2e} annually. The project activity is installed in Monourom village, Bavet City, Svay Rieng Province, the Kingdom of Cambodia.

The project activity generates GHG emission reductions by generate clean form of electricity through renewable solar energy, in the 1st monitoring period from 27-January-2022 to 31-December-2022, the total verified carbon units (VCUs) achieved in the monitoring period are 20,393 tCO_{2e}, which has been

verified by on-site inspection, and checking the related documentations, interview with the project implementer.

The project activity generates GHG emission reductions by replacing the electricity generated by fossil fuel fired power plants in EDC. Thus the project generates GHG emission reductions and produces financial, social and environmental benefits. The project has resulted in the local sustainable development as described in Section 1 of the Joint-PD-MR/1/. The project applied to be registered as a VCS project activity and the 20,393 tCO₂e of emission reductions have been certified as verified carbon units (VCUs) for 1st monitoring period from 27-January-2022 to 31-December-2022.

2 VALIDATION AND VERIFICATION PROCESS

2.1 Method and Criteria

A project specific validation and verification plan was developed to guide the validation and verification auditing process to ensure efficiency and effectiveness.

The purpose of the validation and verification is to present a risk assessment for determining the nature and extent of validation and verification procedures necessary to ensure the risk of auditing error is reduces to a reasonable level. According to the ISO14064-3, the criteria are the policy, procedure or requirement used as reference against which evidence is compared. Therefore, validation of the project description and verification of the monitoring plan and the reported project results were measured for compliance against the following criteria:

- VCS Standard, v4.4/41/
- VCS Program Guide, v4.3/40/
- VCS Program Definitions, v4.3/39/
- VCS-Joint-Project-Description-Monitoring-Report-Template-v4.2/37/

The validation and verification process derived from all items in the validation and verification criteria stated above. Field inspection and techniques based on the project parameters, scope and best professional judgement of the validation and verification team in order to meet a reasonable level of assurance. The validation and verification consisted of the following three phase:

- Document review
- On-site assessment
- The resolution of outstanding issues and the issuance of the final joint validation and verification report and certification.

The validation and verification process derived from all items in the validation and verification criteria stated above. Field inspection and techniques based on the project parameters, scope and best professional judgement of the validation and verification team in order to meet a reasonable level of assurance.

There is no sampling approach utilized by PP and VVB during validation and verification.

2.2 Document Review

The VCS Joint-PD-MR/1/ of the project and supporting background documents related to the project implementation were reviewed. Documents review was conducted to ensure consistency with and identify any deviation from VCS program requirements. Desk review included an examination of the project design details, baseline scenario, additionality, ex ante and monitoring data and parameters, and quantification of GHG emission reductions.

Furthermore, the validation and verification team used additional documentation by third parties like host party legislation, technical reports referring to the project design details, baseline scenario, additionality, monitoring or to the basic conditions and technical data.

The references used in the course of this validation and verification are summarized in Appendix 3.

The validation and verification were performed basing on the documents check and site inspection/measurements, refer to the section 3 and 4 of this report for the validation and verification process detail and corresponding documents review.

2.3 Interviews

Describe the interview process and identify personnel, including their roles, who were interviewed and/or provided information additional to that provided in the Joint-PD-MR/1/ and any supporting documents.

The objective of the interview process was to solicit important information from personnel related to project and relevant to the validation and verification process. Onsite interviews and information discussions were conducted with representatives from project implementer/I1/, Operation Staffs of the solar power plant/I2/, local officers/I3/ and local residents/I4/. The interviews were performed by the validation and verification team on-site (within the project area, in Monourom village, Bavet City, Svay Rieng Province, the Kingdom of Cambodia) and the following is a list of the main interviewees and subject.

No.	Interviewee			Date	Subject, Reference Number/ID	Team member
	Last name	First name	Affiliation			
1.	Bunthi n/I1/	Thon	Green Sustainable Ventures (Cambodia) Co., Ltd. / Site Manager	07-April-2023	- Chronological description of the project activity with documents of key steps of the implementation. - Technical details of the project realization, designing, operational life time - ER calculation - Monitoring system and arrangements - Monitoring device - Calibration - QA/QC - Ownership - local stakeholder consultation - Training - SDG impacts	Lin Shunrong Emanuel Rogier van Mansvelt (Local Expert)
2.	Cham rong /I2/	Seang	Operation staff	07-April-2023	- Technical details of the project realization, designing	
3.	Dara	Khun		07-April-2023		

					- Monitoring system and arrangements - Operation status of the project - Monitoring device - Calibration - QA/QC - Staff competency - Training	
4.	Rathana /13/	Nang	Local Government Officer/ Ministry of Mines and Energy (MME)	07-April-2023	Baseline Scenario	
5.	Sopheak	Som	Local Government Officer/ EDC grid	07-April-2023	Local Stakeholder Consultation issues, SDG impacts, Inputs, Grievances mechanism, Local legislation to renewable energy especially PV	
6.	Sarun /14/	Pich	Local Stakeholders/ Villager	07-April-2023	Local Stakeholder Consultation issues, Local environment impact	
7.	Sarik	Pich		07-April-2023		
8.	Baret	Prom		07-April-2023		
9.	Komsan	Pon		07-April-2023		

2.4 Site Visits

The joint validation and verification site inspection was conducted on 07-April-2023. A ground inspection of the project was conducted during the site visit and the joint validation and verification and VVB interviewed representatives as listed in section 2.3.

During the site inspection, the project site was inspected and documents evidence were checked, details as following table,

Duration of on-site inspection: 07-April-2023				
No.	Activity performed on-site	Site location	Date	Team member
1.	Opening meeting Interview with PP, local officers. Local villagers	Office in solar project plant in Monourom village, Bavet City, Svay Rieng Province, Cambodia	07-April-2023	Lin Shunrong
2	On-site inspection Interview with Operation Staffs	Project Site in Monourom village, Bavet City, Svay Rieng Province, Cambodia	07-April-2023	Emanuel Rogier van Mansvelt (Local Expert)
3	Documents check	Office in solar project plant in Monourom village, Bavet City, Svay Rieng Province, Cambodia	07-April-2023	
4	Finding Summary	Office in solar project plant in Monourom village, Bavet City, Svay Rieng Province, Cambodia	07-April-2023	
5	Close Meeting	Office in solar project plant in	07-April-2023	

		Monourom village, Bavet City, Svay Rieng Province, Cambodia		
--	--	--	--	--

2.5 Resolution of Findings

Material discrepancies identified in the course of the validation and verification are addressed either as CARs, CLs or FARs.

A Corrective Action Request (CAR) is established where:

- The project participants have made mistakes that will influence the ability of the project activity to achieve real, measurable additional emission reductions;
- The VCS Standard Version 4.4 requirements have not been met;
- There is a risk that the emission reductions cannot be monitored or calculated.

A Clarification Request (CL) will be issued where information is insufficient, unclear or not transparent enough to establish whether a requirement is met.

A Forward Action Request (FAR) will be issued when certain issues related to project implementation should be reviewed during the next verification.

A detailed list of the CARs CLs and FAR raised and discussed in the course of this validation and verification is included in Appendix 4 of this report.

In the course of the validation 12 Corrective Action Requests (CARs), 11 Clarification Requests (CLs) were raised and successfully closed. The assessment is included in the report.

2.5.1 Forward Action Requests

This is the joint validation and 1st VCS verification, thus there is no FAR raised previously.

3 VALIDATION FINDINGS

3.1 Project Details

- Project type, scope, and eligibility of the project

Type:

Through the onsite visit, interviewing the project developer and reviewing the project description/1/ and checking the Predesign Studies/5/, Project approval/6/, Business license/4/, it was confirmed that the project is implemented by Green Sustainable Ventures (Cambodia) Co., Ltd., the main purpose of this project activity is to generate clean form of electricity through renewable solar energy. The installed capacity of the project activity is 26MWp, consisting of 57,639 pieces of the solar PV cell modules with unit capacity of 455Wp. The utilization of solar energy from project site is for generation of power, the average annual power delivered to the grid by the project is expected to be 37,085 MWh. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants in Electricite Du Cambodge (EDC) Grid/11/. It's estimated that the project activity could achieve GHG emission reductions of 22,790 tCO₂e annually. The project activity is installed in Monourom village, Bavet City, Svay Rieng Province, the Kingdom of Cambodia.

Sectoral scope:

Therefore, it was confirmed that the applicable VCS sectoral scope is Scope 01: Energy (renewable - / non-renewable).

It was confirmed that the project is not a grouped project.

Eligibility of the project:

The PP has described and justified how the project is eligible under the scope of the VCS Program in Joint-PD-MR as per the section 2.1 of VCS Standard Version 4.4. The assessment is provided as below,

2.1.1

1. The seven Kyoto Protocol greenhouse gases: project activity generates GHG emission reductions including only CO₂ which belong to the six Kyoto Protocol greenhouse gases.
2. Ozone-depleting substances: N/A
3. Project activities supported by a methodology approved under the VCS Program through the methodology approval process: N/A.
4. Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval: The applied methodology ACM0002 (Version 21.0) is approved under CDM Program which is a VCS approved GHG program.
5. Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements: N/A.

2.1.2 “The scope of the VCS Program excludes projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction.”

Via checking the Joint-PD-MR/1/ of the project, it is verified that the project is implemented to install a new green-field solar power plant that generate clean form of electricity through renewable solar energy, which will reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants in Electricite Du Cambodge Grid (EDC), and the additionality is verified as actual, thus there is no assumption of having generated GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction for this project.

2.1.3 “The VCS Program also excludes the following project activities under the circumstances indicated in Table 1”

Via checking the table 1 in VCS Standard Version 4.4, VVB confirmed that the Grid-connected electricity generation activities using wind, geothermal, or solar photovoltaic (PV) power plants in Non-LDCs are excluded, however, via checking the Least Developed Country definition issued by United Nations/27/, VVB confirmed that Cambodia is a Least Developed Country, thus it is concluded that the project does not belong to the project types excluded in Table 1 of VCS Standard Version 4.4.

PP has also described and justified how the project is eligible under the scope of the VCS Program in Joint-PD-MR as per the section 3.1 of VCS Standard Version 4.4. The assessment is provided as below,

1. Via checking the Joint-PD-MR and on-site inspection of the project, CTI confirmed that the project meets all applicable rules and requirements as set out under the VCS Program which has been demonstrated in the whole report, thus the project activities adhere to section 3.1.1 of the VCS Standard.
2. The project applies the CDM approved methodology ACM0002 (version 21.0) which is an eligible VCS methodology along with tool or modules, hence the project activities adhere to section 3.1.2 of the VCS Standard.
3. CDM approved methodology ACM0002 (version 21.0) is verified as the latest version by checking the UNFCCC website, hence the project activities adhere to section 3.1.3 of the VCS Standard.
4. The project complies with all relevant local, regional and national laws, statutes and regulatory frameworks in Cambodia which has been assessed in detail as below, hence the project activities adhere to section 3.1.4 of the VCS Standard.
5. Section 3.1.5 of the VCS Standard is not applicable as no own model have been chosen by the project proponent as all calculation formulas used by PP are derived from the applied methodology ACM0002 (version 21.0)/31/.
6. The project proponent applies the CDM approved methodology ACM0002 (version 21.0) and Tool 01: “Tool for the demonstration and assessment of additionality(version07.0)/32/ for baseline scenario analysis and additionality demonstration which has been assessed in section 3.4.4 and 3.4.5 of this report, hence the project activities adhere to section 3.1.6 of the VCS Standard.
7. Section 3.1.7 of the VCS Standard is not applicable.

8. There is no capacity limit from the Large-scale Consolidated methodology ACM0002 which has been applied by the project.
9. PP will follow the new VCS Program rules as updated by VERRA in the future monitoring period.

In conclusion, the project is eligible to the scope of the VCS Program.

- Project design, including eligibility criteria for grouped projects

The project activity involves the construction and operation of a solar power plant in Monourom village, Bavet City, Svay Rieng Province, the Kingdom of Cambodia owned by Green Sustainable Ventures (Cambodia) Co., Ltd. It has been designed to a single installation of an activity.

This project is not a grouped project.

- Project proponent and other entities involved in the project

The project proponent (Green Sustainable Ventures (Cambodia) Co., Ltd.), and the contact information reported in section 1.5 of the VCS Joint-PD-MR/1/ have been confirmed to be correct through checking the business license of the company/4/, EPC contract/12/, Project Approval/6/, Approval of EIA/10/ and interviewing with the PP representative through the on-site visit/11/.

The other entities involved in the Project (Climate Bridge (Shanghai) Ltd.) and the contact information reported in section 1.6 of the VCS Joint-PD-MR have been confirmed to be correct through checking the business license of the Climate Bridge (Shanghai) Ltd./4/ and interviewing the representative of this company.

- Ownership

The section 1.7 of the Joint-PD-MR has been checked, it is confirmed that the Project Proponent demonstrates that they have the legal right to control and operate project activity.

Via checking the National legal approval to the Environmental Impact Assessment (EIA)/10/, Project Approval/6/, EPC contract/12/, Power Purchase Agreement (PPA)/11/ and the business license of the Green Sustainable Ventures (Cambodia) Co., Ltd./4/, it is verified that the project owner of this solar project is Green Sustainable Ventures (Cambodia) Co., Ltd. and it has the legal right to control and operate the project activity.

By checking Certified Articles of Incorporation and Attested Ownership Structure of the project owner/54/, it is verified that Pestech owned 100% shares of Astoria Solar Farm SDN.BHD who owned 94% equity of project owner. Besides, Salah Essa, a Cambodian individual owned 6% of the equity of project owner. As stated by project owner, aNew Energy, an energy investment expert, introduced Astoria Solar Farm SDN.BHD which is wholly owned by Pestech to establish the project owner company with Salah Essa together.

Furthermore, via checking the Land Lease Agreement/16/ of the project site, it is confirmed that Salah Essa (land owner) who owns the project land has leased the project land to the project owner for 20 years to operate this solar project indicating that the project owner has the land use right to the project land for 20 years.

Thus, it is confirmed that the project owner has the ownership of the project plant, equipment and power generating and carbon credits generated from this project.

- Project start date

The project start date is 27-January-2022 which is verified as the date on which the project began generating GHG emission reductions or removals as per the concept in VCS Standard Version 4.4.

Via checking the Commissioning Letter issued by EDC grid/17/, it is verified that 27-January-2022 is the date when the project plant had been put into operation, hence the start date is correct and in line with the VCS standard.

- Project crediting period

For the VCS crediting period, PP has chosen 21 years renewable crediting period, with first crediting period started from 27-January-2022 to 26-January-2029 which is in line with the concept for crediting period in section 3.9.1 of the VCS Standard Version 4.4.

- Project scale and estimated GHG emission reductions or removals

The average annual ERs of the project are 22,790 tCO_{2e}, which is lower than the threshold of large-projects (300,000 tCO_{2e} per year). Therefore the project falls in the category of “Project” as per definition in VCS standard. And the estimated total Estimated GHG Emission Reductions or Removals during the first 7 years of renewable crediting period (27-January-2022 to 26-January-2029) is 159,532 tCO_{2e}.

- Project description with technologies and measures implemented:

Through the onsite visit, interviewing the project developer and reviewing the project description in Joint-PD-MR/1/, it was confirmed that the project installed a new green-field solar power plant that generate clean form of electricity through renewable solar energy, which will reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants in Electricite Du Cambodge Grid (EDC).

By means of an in-depth review of the Joint-PD-MR and the checks carried out during the on-site visit, an assessment has been carried out whether the project has been implemented and operated in line with the Joint-PD-MR and whether all physical features of the project are in place. The following has been checked: implemented technology, project equipment as well as monitoring equipment.

The verifier has performed a site visit to check the project equipment and interview with operation staffs from power plant, in addition by all the provided evidence, it is found that the project installed a new green-field solar power plant that generate clean form of electricity through renewable solar energy. The total installed capacity is 26MWp which consists of 57,639 pieces of the solar PV cell modules with unit capacity of 455Wp. The conversion ratio of inverters from direct current (DC) to alternating current (AC) is 1.3, i.e.,

DC/AC=1.3. Thus, the total installed capacity is 20MWac¹ on the side of alternating current. The project lifetime of solar PV cell modules is 25 years.

Details of the project equipment technical parameters are given in table below:

Table 3-1 Project equipment (solar cell modules and inverters) technical parameters

Parameter	Unit	Amount
Solar cell module		
Type	-	Q. PEAK DUO XL-G9.3 455W
Number	-	57,639
Rated Power	Wp	455
Rated Voltage	V	44.34
Rated Current	A	10.15
Open Circuit voltage	V	53.18
Short Circuit current	A	10.65
Efficiency	%	≥20
Lifetime	year	25
Manufacturer	-	Hanwha Q CELLS (Qidong) Co., Ltd.
Inverter		
Type	-	SG250HX
Number	-	120
Rated Output power	kW	225
Rated Output Voltage	V	800
Rated Output frequency	Hz	50/60
Manufacturer	-	Sungrow Power Supply Co., Ltd.

Via site inspection of the main equipment and checking the nameplate of the equipment/15/ and cross-checked with the technical agreements of solar cell modules and inverters/14/, CTI verified that the technical parameters listed in above table are correct and actual.

The project started construction on 13-April-2021 by checking the Construction Permit/13/, and started commissioning on 27-January-2022 which has been confirmed by checking the Commissioning Letter issued by EDC grid/17/ and cross-checked with operation log of the project/20/.

The actual operations during the 1st monitoring period are found in accordance with the descriptions provided in the Joint-PD-MR. There is no deviation / change evidenced during this monitoring period.

This verification covers the period from 27-January-2022 to 31-December-2022 (including both days). 20,393 tCO₂e emission reductions are claimed as VCU during this monitoring period.

• Project location

The project site is located at Monourom village, Bavet City, Svay Rieng Province, the Kingdom of Cambodia. Details of the project's location are given in table below:

¹ 20MWac=26MWp/1.3

Table 3-2: Project Location

Item	Project Location	
Host Country	Cambodia	
Region:	Svay Rieng Province	
Location:	Monourom village, Bavet City	
Geographical Coordinates		
East longitude		North latitude
106 ° 02'49.1"E		11 ° 05'20.2"N

The project location has been clearly provided in section 1.12 of the JPM and the detailed coordinates of the project have been provided which have also been verified by site inspection with GPS device and checking the map and the information is verified as correct.

- Conditions prior to project initiation

Via checking the Joint-PD-MR/1/, it is confirmed that the scenario existing prior to the implementation of the project is the electricity delivered to the EDC grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources. The baseline scenario is the same as the scenario existing prior to the implementation of the project activity, which has been confirmed during the site interview with PP and checking the PPA/11/. Thus it is verified that the baseline scenario is reasonable and correct.

And via site inspection and checking the EPC contract/12/, it is verified that all the equipment involved in the project were newly built and there was no existing equipment prior to the project activity hence no existing equipment in baseline scenario.

- Project compliance with applicable laws, statutes and other regulatory frameworks

The Joint-PD-MR has stated that the project is in the renewable energy power generation industry, this industry is confirmed as in line with the laws and regulations/46/ as listed in the Joint-PD-MR.

Also the project is legally approved by Local government by checking the EIA approval/10/ and Project Approval/6/, which means that the project is in line with the national and local laws and regulations based on the local expertise of the local expert in the audit team. And as stated in the Joint-PD-MR/1/ and confirmed by interview with local officers/12/, CTI verified that the project will be subject to regular inspection by local government during the implementation period to ensure continuous compliance.

In conclusion, the project complies with all relevant local, regional and national laws, statutes and regulatory frameworks in Cambodia.

- Participation under other GHG programs:

-Projects registered (or seeking registration) under other GHG program(s)

The project has neither been registered nor seeking registration under any other GHG programs which has been confirmed via checking the UNFCCC/42/, GS/44/, VCS/43/ and other GHG schemes' website. It is verified that the project is seeking registration only in VCS program/43/.

-Rejection by other GHG programs

The project has neither been rejected by any other GHG programs which has been confirmed via checking the UNFCCC/42/, GS/44/, VCS/43/ and other GHG schemes' website.

- Other forms of credit:

-Emissions trading programs and other binding limits

The project is not part of any emission trading program which has been confirmed via checking the UNFCCC/42/, GS/44/, VCS/43/ and other GHG schemes' website. The net GHG emission reductions during this monitoring period from the project was not used for compliance with emission trading programs or to meet binding limits on GHG emissions. The project activity has not participated under any other GHG programs which has been confirmed via checking the UNFCCC/42/, GS/44/, VCS/43/ and other GHG schemes' website, it is verified that the implementation of the project is consistent with the description in the Joint-PD-MR and the monitoring system is fully functional to generate Verified Carbon Units (VCUs) without any double counting for this monitoring period from 27-January-2022 to 31-December-2022 which has been confirmed by checking the Declaration of no double counting and not involved in other GHG scheme/25/.

-Other forms of environmental credit sought or received and eligible to be sought or received

Via interview with the project developer/11/ and based on above assessment, it is confirmed that the project has not created any other form of environmental credit.

-Issuance of public statement(s) to help prevent Scope 3 emissions double claiming

Based on above assessment, it is confirmed that the project is grid-connected renewable solar power project and the project proponent built and operated the power plant to generate the renewable electricity which is supplied to the EDC grid. There are no any wholesalers, distributors or retailers involved. Therefore, the project proponent promised that the emission reduction generated by the project will only seek VCUs and will not be double counting.

-Email notification of the potential risk of Scope 3 emissions double claiming

N/A

- Sustainable development contributions

Via on-site inspection, checking the evidence provided as below and interview with the project implementer, CTI verified that the project contributes the sustainable development through the following aspects:

1. The project generated renewable energy supplied to the power grid (Goal 7 of UN SDG) – verified by checking the Monthly Operation&Maintenance Report/21/ and Electricity sales&purchase receipts/22/ and Emission Reduction Calculation sheet/2/, it is confirmed that 34,694.918

MWh of power were produced from renewable sources during this monitoring period and since project started operation, which replace the powered electricity in EDC grid which is dominated by fossil fuel power generation, hence the contribution to this SDG has been confirmed and the provisions for monitoring and reporting this goal is verified as reasonable;

2. The project provides job opportunities by implementation of the project (Goal 8 of UN SDG) – verified by interview with operation staffs and checking the Staff List/26/, it is confirmed that 7 job opportunities have been provided for local resident during this monitoring period and since project started operation, hence the contribution to this SDG has been confirmed and the provisions for monitoring and reporting this goal is verified as reasonable;
3. Sequester greenhouse gas and mitigate climate changes (Goal 13 of UN SDG) – 20,393 tCO_{2e} GHG emission reductions have been achieved during this monitoring period and since project started operation, which is verified by checking the Emission Reduction Calculation sheet/2/, hence the contribution to this SDG has been confirmed and the provisions for monitoring and reporting this goal is verified as reasonable.

• Additional information relevant to the project, including:

-Leakage management for AFOLU projects

N/A as project is not an AFOLU project.

-Commercially sensitive information

No commercially sensitive information has been excluded from the public version of the project description.

Based on above assessment and demonstration, it is concluded that the description in the project description is accurate, complete, and the nature of the project is actual and understandable, and it is confirmed that the project has been implemented as described in the project description via site inspection.

CL 01, CL 02, CL 03, CAR 01, CAR 02 and CAR 03 were raised and successfully closed. Refer to Appendix 4 for detail assessment.

3.2 Participation under Other GHG Programs

The project has neither been registered nor seeking registration under any other GHG programs which has been confirmed via checking the UNFCCC/42/, GS/44/, VCS/43/ and other GHG schemes' website. It is verified that the project is seeking registration only in VCS program/43/.

3.3 Safeguards

3.3.1 No Net Harm

In Cambodia, an Environmental Impacts Assessment is required according to National legislation. The EIA was designed by ERD Consultants Co., Ltd./9/ and approved by Ministry of Environment on 30-

November-2021/10/. The analysis of the environmental impacts of the project activity is sufficiently described according to EIA report/9/. The assessment is as below,

It is considered that the environmental impacts and the relevant mitigation methods have been comprehensively described in Section 2.1 and 2.3 of Joint-PD-MR including the topography (land and vegetation), air and water pollution, noise, and solid waste impact on the ecosystem and environment of local area in project site for both construction and operation period according to the EIA report/9/.

Associated with the review of EIA report/9/, there was no adverse effect on the air, water and solid waste, for the project subject to the implementation of pollution mitigation measures according to the EIA report/9/. There was also no any endangered species found in the project location.

As per the requirements from approval of EIA/10/, project owner has planted grass under solar photovoltaic panels to prevent water loss and soil erosion during operation period which has been verified by site inspection.

During the on-site interview, the local residents/13/ were interviewed if there was critical impact to their living environment. It is verified that the project activity does not cause adverse effect to their daily life.

Besides, via interview with local officer and stakeholders, it is confirmed that the implementation of the project will improve local socio-economic development through creating career opportunities and paying taxes.

Furthermore, via site interview with PP and local resident, it is confirmed that the project land is a sugar plantation prior to construction but leased to the PP by the land owner for project implementation which has also been verified by checking the Land Lease Agreement/16/. Via checking the EIA/9/ and the Report of Cambodia Socio-Economic Survey 2021/28/ which show that the major income sources is from wages and salaries, it is confirmed that no employment on plantations is reported among major income sources, thus it is confirmed that the project replaces sugar plantation with little impact on the income of local residents.

Besides, as per the Land Lease Agreement/16/, it is confirmed that land owner purchased the project land from previous landowners with paid land sold fees to owners with all owners' agree with sell and there is no economic dispute existed in the project land. Therefore, it is verified that no negative impacts will be caused by the project to local socio-economic development.

In conclusion, there were some impacts on environment during the construction and operation periods which has been demonstrated in the Joint-PD-MR/1/ and EIA/9/, also the proper measurements have been carried out by project implementer to mitigate such impacts.

CAR 04 was raised and successfully closed. Refer to Appendix 4 for detail assessment.

3.3.2 Local Stakeholder Consultation

Local stakeholder consultation prior the project started construction

Based on the EIA/9/, PP has identified the local stakeholder involved local village chiefs and villagers.

A stakeholder survey was carried out by PP in November 2020 which was prior to the start date of the construction.

PP has entrusted ERD Consultants Co., Ltd. to visit local village chiefs at the project site in November 2020 and collect their comments by asking questions using questionnaire as shown in the Monthly Operation&Maintenance Reports/21/. It is confirmed no comments were received from local village chiefs and they all support the project.

During the construction process, by provided jobs to local villagers, they know the project more and suppose the project can improve their living standards and promote local economic development, and via checking the Health Safety Environment monthly report/23/, it is confirmed that villagers are very supportive of the project and no comments received. Thus no need to change the project design based on the stakeholder inputs.

Mechanism for on-going communication and consultation:

During the implementation stage of this monitoring period, via checking the Monthly Operation&Maintenance Reports/21/ which include the documented community issues through conducted physical outreach to the neighbouring communities and interview with the local officer/12/ and local residents/13/ during site visit, it is confirmed that the PP has conducted follow-up interviews and ongoing communication with local stakeholders to collect any comments or inputs to the project implementation.

And by checking the Monthly Operation&Maintenance Reports/21/, it is confirmed that there was a record of the comment from local, the nearby Pagoda Monks directly prevented the maintenance team from trimming who want to trim the trees to avoid tripping over the power transmission line. This has been considered by project manager immediately and invited the village chief to discuss with the Monk head and explained the risk of the tree, and finally got the agreement from the local resident as recorded in the Monthly Operation&Maintenance Reports/21/. Except this, no more comments have been received from local stakeholders.

Moreover, the project owner carried out a face-to-face survey for the local stakeholder to collect the relevant comments and suggestions in February 2022 during this 1st monitoring period. Via checking the records of the survey/24/, interview with local stakeholders during site visit, it is verified that all the respondents believe that the Project will not have any negative environmental impact. It is verified that all local stakeholders who participated the survey were satisfied with the implementation of the project which has been confirmed by site interview with local stakeholders.

Via interview with the local officer/13/ and local residents/14/ during site visit, it is verified that there were no comments and issues from the local stakeholders during the implementation period and the project passed all the requirements from local government. Hence there is no need to change the project design based on the stakeholder inputs.

Furthermore, via site inspection, CTI confirmed that a grievance signage was put at the project gate so that to collect the grievances or comments from local stakeholders anytime, and via checking the signage, CTI verified that there was no any grievances or comments received during this monitoring period.

CL 04, CL 05 and CAR 05 were raised and successfully closed. Refer to Appendix 4 for detail assessment.

3.3.3 Environmental Impact

In Cambodia, an Environmental Impacts Assessment is required according to National legislation. The EIA was designed by ERD Consultants Co., Ltd./9/ and approved by Ministry of Environment on 30-November-2021/10/. The analysis of the environmental impacts of the project activity is sufficiently described according to EIA report/9/. The assessment is as below,

It is considered that the environmental impacts and the relevant mitigation methods have been comprehensively described in Section 2.3 of Joint-PD-MR including the topography (land and vegetation), air and water pollution, noise, and solid waste impact on the ecosystem and environment of local area in project site for both construction and operation period according to the EIA report/9/.

The environmental impacts of the project were sufficiently assessed by means of an Environmental Impact Assessment (EIA) according to the “Law on Environmental Protection and Natural Resource Management”/46/. Every aspect of environmental impact has been considered in the EIA report with corresponding measures during project construction and operation.

During the construction period, as per the EIA/9/, it is confirmed that the potential environmental impacts have been sufficiently identified for the construction period, such as topography (land and vegetation), air and water pollution, noise, and solid waste etc. The conclusion of the analysis has been described in the Joint-PD-MR as per the EIA report/9/ and the related mitigation measures have been listed as per the approval of EIA report/10/. Due to the project has been implemented during the site inspection, it is confirmed that the impact during the construction period has been solved, the project owner has planted grass under solar photovoltaic panels to prevent water loss and soil erosion which has been verified by site visit, and no significant environmental impacts are expected from the project activity.

Associated with the review of EIA, there was no adverse effect on the topography (land and vegetation), air and water pollution, noise, and solid waste for the project subject to the implementation of pollution mitigation measures according to the EIA report. There were no exhaust gas emissions, no domestic wastewater and solid waste observed by site inspection but noise heard which is generated by inverters and transformers, however, noise level measurements has been conducted to ensure that noise level is complied with International Finance Corporation Environmental, Health, and Safety Guidelines/29/. And by site inspection, it is confirmed that the lowest distance from project site to villager’s home is more than 1km, and as confirmed by the local resident who lived in that village, it is verified that no noise contribution to the surrounding boundary after distance attenuation. Hence VVB verified that noise has little influence on the boundary acoustic environment which will not impact the local residents.

In conclusion, the environmental impact during the project construction and operation period will be minor. Via site inspection and interview with local residents, it is verified that PP has taken appropriate measures to minimize adverse environmental impacts and the measures are effective. And no negative comments were received from local resident to the project implementation.

CL 06 was raised and successfully closed. Refer to Appendix 4 for detail assessment.

3.3.4 Public Comments

The project sought the registration with the VCS Program, according to VCS requirements the Project Description was submitted for public comment period. The project was allocated with VCS ID 4036. The public comment period started on 17-January-2023 and ended on 16-February-2023 (<https://registry.verra.org/app/projectDetail/VCS/4036>).

No comments were received during this period which has been verified by checking the dedicated website as above.

3.3.5 AFOLU-Specific Safeguards

N/A

3.4 Application of Methodology

3.4.1 Title and Reference

The approved consolidated methodology applied in the project activity is ACM0002 – “Grid-connected electricity generation from renewable sources” (version 21.0)/31/.

This methodology also refers to the latest approved version of the following tools and guidelines:

Tool 01 Tool for the demonstration and assessment of additionality (Version 07.0.0)/32/

Tool 07 Tool to calculate the emission factor for an electricity system (version 07.0)/34/

Tool 05 Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (Version 03.0)/33/

Tool 24 Common practice (Version 03.1)/35/

Tool 27 Investment analysis (Version 12.0)/36/

CAR 06 was raised and successfully closed. Refer to Appendix 4 for detail assessment.

3.4.2 Applicability

Demonstration regarding the applicability of the methodology, and any tools selected by the project proponent as provided in below table.

Table 3-3 Assessment of Applicability conditions of ACM0002 (version 21.0)

No.	Applicability conditions	Validation Assessment	Conclusion
1	This methodology applies to project activities that include retrofitting, rehabilitation (or refurbishment), replacement or capacity addition of an existing power plant or construction and operation of a Greenfield power plant.	Via site inspection and checking the Predesign Studies/5/, Project approval/6/, EPC contract/12/, CTI verified that this project activity is a Greenfield power plant. Hence this criteria is applicable for this project activity.	Applicable
2	This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant.	Via site inspection and checking the Predesign Studies/5/, Project approval/6/, EPC contract/12/, CTI	Applicable

No.	Applicability conditions	Validation Assessment	Conclusion
	(b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plants/units. (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s).	verified that this project activity is a Greenfield power plant. Hence this criteria is applicable for this project activity.	
3	In case the project activity involves the integration of a Battery Energy Storage System (BESS), the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s).	Via site inspection and checking the Predesign Studies/5/, Project approval/6/, EPC contract/12/, CTI verified that this project activity is a Greenfield solar power plant, no integration of a Battery Energy Storage System (BESS) involved. Hence this criteria is N/A.	N/A
4	The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity;	Via site inspection and checking the Predesign Studies/5/, Project approval/6/, EPC contract/12/, CTI verified that this project activity is a Greenfield solar power plant, no capacity additions, retrofits, rehabilitations or replacement and integration of a Battery Energy Storage System (BESS) involved. Hence this criteria is N/A.	N/A

No.	Applicability conditions	Validation Assessment	Conclusion
	(c) In case of Greenfield project activities applicable under paragraph 5 (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g. by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.		
5	In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply.	Via site inspection and checking the Predesign Studies/5/, Project approval/6/, EPC contract/12/, CTI verified that this project activity is a Greenfield solar power plant but not a hydro power plant. Hence this criteria is N/A.	N/A

No.	Applicability conditions	Validation Assessment	Conclusion
	(i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity. (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project		
6	In the case of integrated hydro power projects, project proponent shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	Via site inspection and checking the Predesign Studies/5/, Project approval/6/, EPC contract/12/, CTI verified that this project activity is a Greenfield solar power plant but not a hydro power plant. Hence this criteria is N/A.	N/A
7	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units;	Via site inspection and checking the Predesign Studies/5/, Project approval/6/, EPC contract/12/, CTI verified that this project activity is a Greenfield solar power plant but not involved in any of the given criteria. Hence this criteria is N/A.	N/A
8	In the case of retrofits, rehabilitations, replacements, or capacity additions, this	Via site inspection and checking the Predesign Studies/5/, Project approval/6/, EPC contract/12/, CTI	N/A

No.	Applicability conditions	Validation Assessment	Conclusion
	methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	verified that this project activity is a Greenfield solar power plant but not involved in any of the given criteria. Hence this criteria is N/A.	
Conclusion: Methodology ACM0002 (version 21.0) is applicable to the project activity.			

In addition, the project meets the applicability conditions of the applied tools as follow table,

Table 3-4 Assessment of Applicability conditions of methodological tool

Tool	Criteria	Validation Assessment	Conclusion
(a) Tool 01 Tool for the demonstration and assessment of additionality (Version 07.0)	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	The methodology selected for the proposed project requires the use of this tool. Hence this criteria is applicable.	Applicable
	Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	The methodology selected for the proposed project requires the use of this tool. Hence this criteria is applicable.	Applicable
Tool 05 Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (Version 03.0)	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is	Via site inspection and checking the PPA/11/, it is confirmed that the electricity consumption of the project will be provided by the EDC grid, which falls under scenario A. Hence this criteria is applicable.	Applicable

Tool	Criteria	Validation Assessment	Conclusion
	either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.		
	This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	Via site inspection and checking the PPA/11/, it is confirmed that the electricity is supplied to the EDC grid, which falls under scenario I. Hence this criteria is applicable.	Applicable

Tool	Criteria	Validation Assessment	Conclusion
	This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO ₂ emissions.	Via site inspection, it is confirmed that no captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. Hence this criteria is N/A.	N/A
Tool to calculate the emission factor for an electricity system (Version 07.0)	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	Via site inspection and checking the PPA/11/, it is confirmed that the electricity is supplied to the EDC grid, hence this tool can be used to calculate the values of OM, BM and CM of this project. Hence this criteria is applicable.	Applicable
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and	The conservative value of Default Emissions Factor (DEF) issued by IFI Technical Working Group (IFI TWG)/51/ has been used as the emission factor for the grid of the project. Via checking all the public source of the Emissions Factor value, it is confirmed that the latest available version that can be obtained before the validation (in line with the requirement of the CDM Tool 07: “Tool to calculate the emission factor for an electricity system” (Version 07.0)) is IFI Dataset of Default Grid Factors v.3.2 which is published in April 2022/51/. So the data source is verified in line with the Tool 07 request. Hence this criteria is applicable.	Applicable

Tool	Criteria	Validation Assessment	Conclusion
	not to other aspects such as transmission capacity.		
	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Via checking the project location by site inspection, it is confirmed that there is no part of the power system of this project located in Annex I countries. Hence this criteria is N/A.	N/A
	Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	The project is a solar power project, no biofuels involved. Hence this criteria is N/A.	N/A
Tool 27 Investment analysis (Version 12.0)	This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario	As above demonstration, the project applies the methodological tool “Tool for the demonstration and assessment of additionality”. Hence this criteria is applicable for this project activity.	Applicable
	In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The applied methodology ACM0002 (Version 21.0) requires the use of this tool to demonstrate the investment analysis of this project. Hence this criteria is applicable for this project activity.	Applicable
Tool 24 Common practice (Version 03.1)	This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Tool for the demonstration and assessment of additionality”, or baseline and	As assessed above, the project applies the methodological tool “Tool for the demonstration and assessment of additionality” for the demonstration of additionality, the project can use the common practice	Applicable

Tool	Criteria	Validation Assessment	Conclusion
	monitoring methodologies that use the common practice test for the demonstration of additionality.	test for the demonstration of additionality. Hence this criteria is applicable for this project activity.	
	In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	Via checking the applied methodology, CTI confirmed that the methodology defines approaches for the conduction of the common practice test that are same to those described in this methodological tool. Hence this criteria is applicable for this project activity.	Applicable
Conclusion: All the relevant methodological tools are applicable to the project activity.			

CL 07 was raised and successfully closed. Refer to Appendix 4 for detail assessment.

3.4.3 Project Boundary

The project boundary basically defines the physical and geographical boundary of the project facility and it is well defined in the Joint-PD-MR/1/ according to ACM0002 Grid-connected electricity generation from renewable sources (version 21.0)/31/. The project boundary includes following,

Project boundary has been defined in the JPM according to the applied methodology ACM0002/31/ as includes the project power plant/unit and all power plants/units connected physically to the EDC grid that the project power plant is connected to which is verified by checking the PPA of the project/11/and on-site inspection.

Via site inspection and checking the PPA of the project/11/ and the technical flow chart shown in the project site/18/, it is verified that project boundary is clearly defined in the JPM as per the methodology.

Emissions sources included in the project boundary have been appropriately included in the JPM. CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity is covered for baseline scenario and no emission source in the project scenario for solar photovoltaic project which is confirmed in line with the methodology ACM0002/31/.

It is concluded that the project boundary and selected sources are correctly justified for the project.

3.4.4 Baseline Scenario

The PP has applied an approved baseline and methodology ACM0002, Version 21.0 which is approved under CDM scheme.

And as per the applied methodology, the baseline scenario for greenfield power plant has been defined as “electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in TOOL07”.

Hence PP has correctly defined the baseline scenario in the JPM in line with the applied methodology

3.4.5 Additionality

The JPM applies the stepwise approach as given by section 5.2 of the ACM0002/31/ and "Tool for the demonstration and assessment of additionality" (Version 07.0)/32/.

Step 1: Identification of alternative to the project activity consistent with current laws and regulations

Step 1a: Define alternative scenarios to the project activity

The alternatives available to the project participant or similar project developers that provide outputs or services comparable are identified as below

Alternatives	VVB Assessment
(a) The proposed project activity undertaken without being registered as a VCS project activity	It is an appropriate alternative which confirmed in line with the applied methodology and tool.
(b) Continuation of the current situation (no project activity or other alternatives undertaken)	It is an appropriate alternative which confirmed in line with the applied methodology and tool.

No other alternatives need to be identified as per the requirement of the applied methodology.

Step 1b: Consistency with mandatory laws and regulations

Via searching the public website with laws and regulations in Cambodia, it is confirmed that there is no legal prohibition regarding the implementation of the project currently or in the future by checking the Cambodia Basic Energy Plan/46/, and continuation of the current situation is that electricity delivered to EDC grid is still in line with the Electricity Law of The Kingdom of Cambodia/46/ and no laws or regulations in Cambodia that prohibit the power grid from providing the same annual electricity generation.

As assessed above, CTI confirmed that the above alternatives (a) and (b) are considered to follow all mandatory applicable legal and regulatory requirements which are verified based in validation team’s local expertise. The outcome of Step 1b is same to Step 1a.

Step 2: Investment analysis

Sub-step 2a: Determine appropriate analysis method

According to "Tool for the demonstration and assessment of additionality" (Version 07.0)/32/, if the project activity and the alternatives identified in Step 1 generate no financial or economic benefits other than carbon related income, then apply the simple cost analysis (Option I). Otherwise, use the investment comparison analysis (Option II) or the benchmark analysis (Option III).

For the project, the alternative to the project activity is the supply of electricity from a grid which is not considered as an investment by project participant, so the benchmark analysis (Option III) is used, which is also in line with the request of the applied methodology.

Sub-step 2b: Option III. Apply benchmark analysis

As per the tool, PP has chosen IRR as the financial/economic indicator, and as per the section 38 (b) of the tool, the local commercial lending rates can be the benchmark, so the local lending rate offered by National Bank of Cambodia (NBC)/30/ is chosen by the PP for investment analysis which is verified as reasonable and in line with the request of the Tool 01. Furthermore, investment decision defined date of the project is the date on signing the PPA with the EDC grid/11/, i.e., 27-August-2019, hence Cambodian average commercial bank prime lending rate 2018 which is 10.61% is verified as the correct benchmark for the project investment analysis.

Sub-step 2c: Calculation and comparison of financial indicators

Via checking the Financial investment analysis sheet (data source of the inputs value for IRR calculation) /7/, CTI confirmed that the Financial investment analysis sheet which confirmed as including prior consideration of carbon revenue was completed in May 2019, besides, investment decision defined date of the project is the date on signing the PPA with the EDC grid/11/, i.e., 27-August-2019 which can be defined as the investment decision date, hence it is verified that the time interval between Financial investment analysis sheet/7/ and investment decision made by PP is less than half year, therefore CTI confirmed that input values from Financial investment analysis sheet/7/ are valid and applicable at the time of the investment decision.

Furthermore, via checking all the report evaluation standards listed in the Financial investment analysis sheet/7/, especially for all the financial input values which has been assessed in below table, VVB verified that the values are all correctly provided in the Financial investment analysis sheet/7/. Hence based on the local expertise of the local expert, it is verified that all the input values from Financial investment analysis sheet/7/ are valid due to the it was completed by a professor who has most renewable energy experience and enjoys developing an options pool (structuring, financing, technology) from the third party aNew Energy, the professor prepared the Financial investment analysis sheet following all the Cambodian tax regulations, standards and laws/47/. Refer to below table for detail assessment to each input value.

Parameter	Input values	Source	VVB Assessment
Total static investment	24.718 Million USD	Financial investment analysis sheet	The value has been derived from Financial investment analysis sheet/7/. The Total static investment of the project including engineering, construction and equipment, etc.

			Due to the project has been operated during the validation process, via checking the evidence of Investment details sheet & consulting fee/8/, CTI confirmed that the actual investment costs of the project is 24,595,722 USD, which is 99% of the estimated value of 24,718,000 USD in Financial investment analysis sheet/7/. Therefore, CTI confirmed that the values in Financial investment analysis sheet/7/ is reasonable and conservative.
O&M cost	Annual average: 398,972 USD	Financial investment analysis sheet	The Annual O&M cost is derived from the Financial investment analysis sheet/7/. It encompasses sub-items of Maintenance fees (clean and security), Salary and welfare, Insurance fee and Land lease fee. The maintenance fee is calculated as 0.5% of the Fixed assets. The actual O&M operation has been conducted by a third company, and via checking the signed “GSV PCL O&M Contract (Executed)"/55/, it is confirmed that the actual annual maintenance fee together with salary and welfare is 157,000USD which is higher than the estimated maintenance fee together with salary and welfare (123,590 USD+28,728USD=152,318USD), hence the estimated maintenance fee is reasonable. There are total 7 staffs which is verified by cross-checking the Staff List/26/. Salary is 3,600 USD/person/year which is confirmed as reasonable by checking the “GSV PCL O&M Contract (Executed)"/55/ and is higher than the average level by cross-checking the public data of the average salary level in Cambodia/56/ that showed the average annual level for Bachelor education level (most of staff are Bachelors) is 23,300USD. The welfare is calculated as 14% of the annual salary which is confirmed as reasonable by checking the “GSV PCL O&M Contract (Executed)"/55/. Refer to above assessment to maintenance fee for reasonability of also salary and welfare. Land lease fee is 150,000USD/year which is same to fee given in Lease Agreement/16/. Insurance fee is calculated as 0.3% of the Fixed assets. This fee is considered as not main parts of the O&M cost and the values are confirmed as reasonable by checking the Insurance list and related Insurance policy/57/. Furthermore, if the O&M Costs decrease by 132.48%, the IRR of project activity reach the benchmark. However, it is not likely to implement the project activity without the O&M cost to make the project activity IRR to cross the benchmark.
Annual electricity	Average: 37,544 MWh	Financial investment	The value is derived from the Financial investment analysis sheet/7/.

delivered to the grid		analysis sheet	CTI confirmed that the Annual electricity delivered to the grid in the first year is calculated as 39,359 MWh based on the installed capacity of 26 MWp and annual photovoltaic electricity yield of 1,513.8 kWh/KWp which has been verified as reasonable by cross-checking the photovoltaic electricity yield data from Global Solar Atlas website/48/. Besides, by checking the technical agreements of solar cell modules provided by manufacturer/14/, it is confirmed that the annual degradation rate of solar cell modules is 0.5%, the annual electricity delivered to the grid during the whole lifetime is calculated as the electricity delivered to the grid last year minus the electricity reduced due to the degradation of solar PV module, which means that the Annual electricity delivered to the grid will be decreased yearly in the project lifetime. If the Annual electricity delivered to the grid increases by 17.83%, the IRR of project activity reach the benchmark. Based on above assessment, it is impossible for Annual electricity delivered to the grid increases by 17.83% to make the project activity IRR to cross the benchmark.
Electricity tariff (without VAT)	76 USD/MWh	Financial investment analysis sheet	The value is derived from Financial investment analysis sheet/7/. Via checking the PPA/11/ and actual sales receipts of electricity/22/ for year 2022, it is confirmed that the Electricity tariff is fixed as 76 USD/MWh which is same to the one used in financial investment analysis. And due to the PPA/11/ was signed with the grid which has ensure the Electricity tariff is fixed as 76 USD/MWh, so it is confirmed that the project participant has use the highest possible tariff that they may receive by supplying the electricity to the grid to assess the economic attractiveness of the project activity which is verified in line with the request of section 34 of the applied methodology. If the Electricity tariff increases by 17.83%, the IRR of project activity reach the benchmark. Based on above assessment, it is impossible to increase Electricity tariff by 17.83% to make the IRR to cross the benchmark.
Installed capacity	20 MW (26MWp)	Financial investment analysis sheet	The value is derived from Financial investment analysis sheet/7/. Via checking the Project approval/6/, EIA approval/9/, EPC contract /12/ and Nameplate of the equipment /15/, it is verified that the value of installed capacity is correct.
Average annual operating hours	1,877 Hour	Financial investment analysis sheet	The value is derived from Financial investment analysis sheet/7/. The value is verified as reasonable by cross-checking the photovoltaic electricity yield data from Global Solar Atlas website/48/.

			Furthermore, by comparing with the average annual operating hour data of another VCS registered solar project in Cambodia (VCS 3613), the figure of the project is lower than 3613 case with average annual operating hour of 1,971 hour. Hence, it is confirmed that the value is reasonable.
Operation period	20 years	Financial investment analysis sheet	The lifetime of the project is defined as 20 years in Financial investment analysis sheet/7/ and cross checked with the lease agreement/16/ and PPA/11/. The value is correct and consistent.
Depreciation period	20 years	Financial investment analysis sheet	The Depreciation period is defined as 20 years in Financial investment analysis sheet/7/ and cross checked with the Cambodian 2018 Tax Booklet/47/. Via checking the Booklet/47/, the annual depreciation charges can be calculated on the useful life by using the straight-line method. Since the useful life of this solar project is 20 years confirmed in the PPA/11/, the depreciation period is 20 years is verified as correct and reasonable.
Depreciation rate	5%	Financial investment analysis sheet	The Depreciation rate is defined as 5% in Financial investment analysis sheet/7/ and cross checked with the Cambodian 2018 Tax Booklet/47/. The value is correct and consistent.

Via checking the calculation formula and calculation process of the value of pre-tax project Internal Rate of Return (IRR) in the calculation sheet/3/, CTI confirmed that the formula and inputs values are correct and actual by checking the Financial investment analysis sheet/7/ for each input value, thus CTI verified that the calculation results of the IRR for project scenario is correct.

And by checking the calculation process of the value of IRR in the calculation sheet/3/, CTI confirmed that the IRR of the project activity without VCU revenues is 7.88%, and the IRR of the project activity without VCU revenues is lower than the value of benchmark 10.61% which means the project activity is not financial attractive.

By considering the carbon revenue, the IRR can reach 9.09% which is higher than the IRR of the project activity without VCU revenues, but still less than the benchmark, but by considering the continuous revenue from the carbon and the return on investment for the project with carbon revenues can be improved to a certain extent, PP decided to apply VCS when decide to investment the project which has been confirmed by site interview with PP.

Sub-step 2d: Sensitivity analysis

As per the Tool 01 and Investment Analysis (version 12.0), a sensitivity analysis has been provided in the Joint-PD-MR and the calculation sheet/3/. The analysis is assessed as follow,

The sensitivity analysis was demonstrated through two manners:

1. Varying $\pm 10\%$ of four critical parameters (Total static investment, Annual O&M cost, Annual electricity delivered to the grid and Electricity tariff). The selection is checked as in line with the requirements in Investment analysis (version 12.0) of “Only variables, including the total static investment, that constitute more than 20% of either total project costs or total project revenues should be subjected to reasonable variation”, the Total static investment and Annual electricity delivered to the grid and Electricity tariff are checked as constitutes more than 20% of total project costs. The total O&M throughout the project lifetime is checked as accounts for more than 20% of the project cost.

Via checking the sensitivity analysis for these four critical parameters as provided in Joint-PD-MR and the calculation sheet/3/, CTI verified that the critical point that the project activity becomes more financial attractive than uncovered anaerobic lagoon by varying $\pm 10\%$ of three critical parameters.

2. Threshold analysis by varying the above four parameters to make the IRR of project activity reach the benchmark. The threshold analysis of each parameter is assessed individually by the audit team as below,
 - a) If the Total static investment decreases by 16%, the IRR of project activity reach the benchmark, however, due to the project has been operated during the validation process, via checking the evidence of Investment details sheet & consulting fee/8/, CTI confirmed that the actual investment costs of the project is 24,595,722 USD, which is 99% of the estimated value of 24,718,000 USD in Financial investment analysis sheet/7/. Therefore, CTI confirmed that the values in Financial investment analysis sheet/7/ is reasonable and conservative and it is impossible to decrease total static investment by 16.00% to make the IRR to cross the benchmark.
 - b) If the Annual electricity delivered to the grid increases by 17.83%, the IRR of project activity reach the benchmark. Annual electricity delivered to the grid in the first year is calculated as 39,359 MWh based on the installed capacity of 26 MWp and annual photovoltaic electricity yield of 1,513.8 kWh/KWp which has been verified as reasonable by checking the photovoltaic electricity yield data from Global Solar Atlas website/48/. Besides, by checking the technical agreements of solar cell modules provided by manufacturer/14/, it is confirmed that the annual degradation rate of solar cell modules is 0.5%, the annual electricity delivered to the grid during the whole lifetime is calculated as the electricity delivered to the grid last year minus the electricity reduced due to the degradation of solar PV module, which means that the Annual electricity delivered to the grid will be decreased yearly in the project lifetime, so it is impossible for Annual electricity delivered to the grid increases by 17.83% to make the project activity IRR to cross the benchmark.
 - c) If the O&M Costs decrease by 132.48%, the IRR of project activity reach the benchmark. However, it is not likely to implement the project activity without the O&M cost to make the project activity IRR to cross the benchmark.
 - d) If the Electricity tariff increases by 17.83%, the IRR of project activity reach the benchmark. However, due to the project has been operated, via checking the PPA/11/ and actual sales receipts of electricity/22/ for year 2022, it is confirmed that the Electricity tariff is fixed as 76 USD/MWh which is same to the one used in financial investment analysis. And due to the PPA/11/ was signed with the grid which has ensure the Electricity tariff is fixed as 76 USD/MWh,

so it is confirmed that the project participant has use the highest possible tariff that they may receive by supplying the electricity to the grid to assess the economic attractiveness of the project activity which is verified in line with the request of section 34 of the applied methodology. Hence, in conclusion, it is impossible to increase Electricity tariff by 17.83% to make the IRR to cross the benchmark.

In conclusion, the investment analysis concludes that the project IRR (pre-tax) will not reach the benchmark of 10.61% and the project activity is unlikely to be financially attractive. Threshold analysis is further proved this. The sensitivity analysis and threshold analysis was reproduced by the validation team and evaluated to be correct. Based above, it can confirm that the financial unattractiveness of the project is robust and thus the alternative (b) is the plausible baseline scenario.

Hence the alternative (b) is considered as baseline scenario which is “Continuation of the current situation (no project activity or other alternatives undertaken)”.

Based on the checking the data provided in Joint-PD-MR and above related assessment, it proves that the baseline scenario determined in the Joint-PD-MR is correct and reasonable. Therefore, baseline scenario is identified transparently for the project activity.

The assessment team has reviewed the Joint-PD-MR in line with the applied methodology and methodological tool and CTI confirmed that PP has correctly identified the baseline scenario.

Step 3: Barrier analysis

PP does not employ the barrier analysis which is confirmed in line with the tool.

Therefore, proceed to Step 4.

Step 4: Common practice analysis

The common practice analysis was checked strictly followed Methodological tool “Common practice” (Version 03.1)/44/.

Sub-Step 4a-1: calculate applicable capacity or output range as +/-50% of the design capacity or output of the proposed project activity

The install capacity of the project is 26 MWp (20 MWac), so the applicable capacity range as +/-50% of the total design capacity is 13 ~ 39 MWp (10 ~ 30 MWac).

Sub-Step 4a-2: identify similar projects (both CDM and non-CDM) which fulfill all of the following conditions:

- a) *The projects are located in the applicable geographical area,*
- b) *The projects apply the same measure as the proposed project activity,*
- c) *The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity,*

- d) *The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g. clinker) as the proposed project plant,*
- e) *The capacity or output of the projects is within the applicable capacity or output range calculated in Step 1,*
- f) *The projects started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of proposed project activity, whichever is earlier for the proposed project activity.*

Demonstration as follow,

- a) *The applicable geographical area selected for common practice of Cambodia is assessed as appropriate and reasonable as the whole country has been considered.*
- b) *Same measure is defined as: uses solar to generate electricity*
- c) *Same energy source/fuel and feedstock: solar energy as energy source/fuel and feedstock*
- d) *Comparable quality: Solar power plant and produce electricity technology*
- e) *Output range: total installed capacity from 13 MWp to 39 MWp (10 ~ 30 MWac)*
- f) *Commercial operation started before 27-January-2022.*

Thus, the solar power plants of the capacity from 13 MWp (10 MWac) to 39 MWp (30 MWac) operational before 27-January-2022 in Cambodia are identified as similar projects.

The information source from EDC grid website/49/ for the common practice analysis is available and checked by the audit team. The information used is evaluated to be credible.

Via the source, CTI confirmed that there are no similar projects identified in Cambodia based on the above criteria.

Step 3: within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number N_{all} .

There are no similar projects identified.

Hence $N_{all}=0$.

Step 4: within similar projects identified in Step 3, identify those that are different to the technology applied in the proposed project activity. Note their number N_{diff} .

Due to $N_{all}=0$, $N_{all}=N_{diff}=0$.

Step 5: calculate factor $F=1-N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

For this project, $F=1-N_{diff}/N_{all}=1-1=0$ and $N_{all}-N_{diff}=0$.

It can be concluded that the project is not a common practice.

In conclusion, the project meets the criteria and tool “Tool for the demonstration and assessment of additionality (Version 07.0)”, thus deemed as additional.

Besides, as per the Section 3.14.1 of VCS Standard (Version 4.4), PP has demonstrated regulatory surplus in JPM which has been checked by VVB as below,

Via searching the public website with laws and regulations in Cambodia, it is confirmed that there is no legal law and regulation to mandate to implement solar energy for electricity generation in Cambodia by checking the Electricity Law of The Kingdom of Cambodia, Long-term strategy for Carbon Neutrality, Cambodia Basic Energy Plan/46/.

Furthermore, via site interview with local officers, it is also confirmed that using renewable energy for electricity generation are encouraged by the state and not mandatory. Therefore, the project activity is not mandated by any law, statute, or other regulatory framework, it is deemed as regulatory surplus.

CL 08, CL 09, CL 10, CAR 07 and CAR 08 were raised and successfully closed. Refer to Appendix 4 for detail assessment.

3.4.6 Quantification of GHG Emission Reductions and Removals

For validation of the estimated GHG emission reductions in the Joint-PD-MR/1/ and ER calculation sheet/2/,

Further, the VVB has downloaded from the UNFCCC website the applicable version of the CDM methodology,

Via verify the Joint-PD-MR/1/, it is confirmed the calculation of ERs is done as per the applied methodology (ACM0002 ver. 21.0)/31/ with follow steps listed below.

Baseline emissions

Via checking the paragraph 47 of the applied methodology, the baseline emissions BE_y in a year y are calculated as:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y} \quad (1)$$

Where:

BE_y	=	Baseline emissions in year y (t CO ₂ /yr)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM-project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	=	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using TOOL 07 (t CO ₂ /MWh)

Via site inspection, it is confirmed that the project is a Greenfield solar power plant, then as per the paragraph 49 of the applied methodology

$$EG_{PJ,y} = EG_{facility,y} \quad (2)$$

Where:

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr).

For $EF_{grid,CM,y}$

Following ACM0002 (Version 21.0), the baseline emission factor should be calculated as combined margin emission factor of EDC grid ($EF_{grid,CM,y}$), consisting of the combination of operating margin emission factor of EDC grid ($EF_{grid,OM,y}$) and build margin emission factor of EDC grid ($EF_{grid,BM,y}$) according to the six steps defined in the “Tool to calculate the emission factor for an electricity system” (version 7.0)/34/.

However, via site interview with the local officer from EDC grid and checking the public data in Cambodia including information source from EDC grid website/49/, CTI confirmed that either the specific fuel consumption of the EDC grid system in most recent five years or CO₂ emission factor of each plant are not available, which lead to the combined margin emission factor ($EF_{grid,CM,y}$) cannot be calculated by the latest version “Tool to calculate the emission factor for an electricity system”/34/.

In addition, there is no available latest emission factor of the EDC grid approved by the relevant National Authority or Designated National Authority (DNA) in Cambodia confirmed by checking the DNA website/50/.

So the UNFCCC website for solving this problem has been searched. The conservative value of Default Emissions Factor (DEF) issued by IFI Technical Working Group (IFI TWG)/51/ has been used as the emission factor for the grid of the project.

Via checking all the public source of the Emissions Factor value, it is confirmed that the latest available version that can be obtained before the validation (in line with the requirement of the CDM Tool 07: “Tool to calculate the emission factor for an electricity system” (Version 07.0)) is IFI Dataset of Default Grid Factors v.3.2 which is published in April 2022/51/. So the data source is verified in line with the Tool 07 request.

By checking the UNFCCC website, for the purpose of promoting greater harmonization, the IFI Technical Working Group (IFI TWG) on GHG accounting maintains a common dataset containing Default Emissions Factor (DEF) of the country’s electricity grid including in-country interconnected grids. The DEFs apply to electricity generation in a country and currently do not consider the impact of interconnections with neighbouring countries.

According to the public “Methodological Approach for the Common Default Grid Emission Factor Dataset”, the common dataset containing DEFs is constructed using a Combined Margin (CM) for the grid that is calculated by combining an Operating Margin (OM) and a Build Margin (BM) as per CDM Tool 07: “Tool to calculate the emission factor for an electricity system” (Version 07.0). This calculation method of CM is in line with ACM0002 (Version 21.0). Besides, the country specific information that is used to calculate OM and BM is derived from The International Energy Agency’s (IEA) energy statistics database. Therefore, the default emission factor from IFI Dataset is verified as accurate and authoritative.

Based on the above mentioned source, it is verified that the latest $EF_{grid,CM,y}$ of solar project and firm energy project (e.g. Hydro, geothermal) in Cambodia in 2021 is 0.8743 tCO_{2e}/MWh and 0.5878 tCO_{2e}/MWh, respectively.

PP has selected 0.5878 tCO_{2e}/MWh for calculate emission reduction for the project which is verified as conservative.

Furthermore, VVB has searched another data source and assessed as below,

a. IGES List of Grid Emission Factors version 10.12/52/. As per this list, there is only 2009 Phnom Penh grid emission factors of solar project in Cambodia which is 0.6413 tCO_{2e}/MWh. However the data is not latest and no latest data available as per the list, so this source is confirmed as not applicable for the project.

b. JCM website/53/. By checking the website, it is confirmed that in case the PV system in a proposed project activity in Cambodia is connected to an internal grid but not connected to the national grid, the emission factor can be set to 0.533 tCO₂/MWh as per JCM Approved Methodology KH_AM002. But it is verified that the calculation method of this emission factor is not in line with CDM Tool 07: “Tool to calculate the emission factor for an electricity system” (Version 07.0). So this source is confirmed as not applicable for the project.

Besides, by checking verra website, the emission factor of another registered Cambodia solar project (VCS ID 3613) is 0.7239 tCO_{2e}/MWh which is higher than 0.5878 tCO_{2e}/MWh of emission factor for the project.

Therefore, the emission factor of the project is verified as reasonable and conservative.

For $EG_{facility,y}$

The ex ante data of $EG_{facility,y}$ is calculated as below,

Annual electricity delivered to the grid in the first year is calculated as 39,359 MWh based on the installed capacity of 26 MWp and annual photovoltaic electricity yield of 1,513.8 kWh/KWp which has been verified as reasonable by checking the photovoltaic electricity yield data from Global Solar Atlas website/48/. Besides, by checking the technical agreements of solar cell modules provided by manufacturer/14/, it is confirmed that the annual degradation rate of solar cell modules is 0.5%, the annual electricity delivered to the grid during the whole lifetime is calculated as the electricity delivered to the grid last year minus the electricity reduced due to the degradation of solar PV module.

Hence, as per the above assessment, the annual ex ante estimated BE_y value is calculated as below sheet for the 1st 7 years crediting period,

Table 3-5 Ex ante estimated BE_y

Year	$EG_{facility,y}$ (MWh)	$EF_{grid,CM,y}$ (t CO _{2e} /MWh)	BE _y (t CO _{2e})
	A	B	C=A*B
27-January-2022~31-December-2022	36,555	0.5878	21,487
01-January-2023~31-December-2023	39,176	0.5878	23,027

01-January-2024~31-December-2024	38,980	0.5878	22,912
01-January-2025~31-December-2025	38,785	0.5878	22,797
01-January-2026~31-December-2026	38,591	0.5878	22,683
01-January-2027~31-December-2027	38,398	0.5878	22,570
01-January-2028~31-December-2028	38,206	0.5878	22,457
01-January-2029~26-January-2029	2,721	0.5878	1,599
Total	271,411	0.587	159,532

Project Emissions

Based on the applied methodology, and via site inspection checking the solar power generation project, CTI confirmed that the project emissions $PE_y=0$.

Leakage

Based on the applied methodology, no leakage emissions need to be considered.

Emission reductions

$$ER_y = BE_y - PE_y = BE_y$$

Based on the above assessment of the ex ante determined values, it is verified that the average annual ex ante determined ERs calculated result is 22,790 tCO₂/yr.

The detail of emission reduction calculation is transparently discussed in the Joint-PD-MR/1/. The assessment team has checked the Joint-PD-MR/1/ and ER sheet/2/ for the detail calculation of all the particulars and found it to be correct.

In conclusion, based on above values for each parameter for determine the ex-ante of emission reduction calculation, the CTI verified that results in Joint-PD-MR are correct for the ex-ante ER values during the first 7 years crediting period.

The ER calculation sheet/2/ has been duly checked. Further it has been checked whether the results have been correctly calculated in Joint-PD-MR/1/ for determination of ex-ante ER. The VVB has further checked the Joint-PD-MR/1/ against the latest version of the applicable methodology for consistency.

CAR 09 and CAR 10 were raised and successfully closed. Refer to Appendix 4 for detail assessment.

3.4.7 Methodology Deviations

There is one methodology deviation applied to the project.

Following ACM0002 (Version 21.0), the baseline emission factor should be calculated as combined margin emission factor of EDC grid ($EF_{grid,CM,y}$), consisting of the combination of operating margin emission factor of EDC grid ($EF_{grid,OM,y}$) and build margin emission factor of EDC grid ($EF_{grid,BM,y}$) according to the six steps defined in the “Tool to calculate the emission factor for an electricity system” (version 7.0).

However, via site interview with the local officer from EDC grid and checking the public data in Cambodia including information source from EDC grid website/49/, CTI confirmed that either the specific fuel consumption of the EDC grid system in most recent five years or CO₂ emission factor of each plant are not available, which lead to the combined margin emission factor ($EF_{grid,CM,y}$) cannot be calculated by the latest version “Tool to calculate the emission factor for an electricity system”/34/.

In addition, there is no available latest emission factor of the EDC grid approved by the relevant National Authority or Designated National Authority (DNA) in Cambodia confirmed by checking the DNA website/50/.

So the UNFCCC website for solving this problem has been searched.

The conservative value of Default Emissions Factor (DEF) issued by IFI Technical Working Group (IFI TWG)/51/ has been used as the emission factor for the grid of the project.

Via checking all the public source of the Emissions Factor value, it is confirmed that the latest available version that can be obtained before the validation (in line with the requirement of the CDM Tool 07: “Tool to calculate the emission factor for an electricity system” (Version 07.0)) is IFI Dataset of Default Grid Factors v.3.2 which is published in April 2022/51/. So the data source is verified in line with the Tool 07 request.

By checking the UNFCCC website, for the purpose of promoting greater harmonization, the IFI Technical Working Group (IFI TWG) on GHG accounting maintains a common dataset containing Default Emissions Factor (DEF) of the country’s electricity grid including in-country interconnected grids. The DEFs apply to electricity generation in a country and currently do not consider the impact of interconnections with neighbouring countries.

According to the public “Methodological Approach for the Common Default Grid Emission Factor Dataset”, the common dataset containing DEFs is constructed using a Combined Margin (CM) for the grid that is calculated by combining an Operating Margin (OM) and a Build Margin (BM) as per CDM Tool 07: “Tool to calculate the emission factor for an electricity system” (Version 07.0). This calculation method of CM is in line with ACM0002 (Version 21.0). Besides, the country specific information that is used to calculate OM and BM is derived from The International Energy Agency’s (IEA) energy statistics database. Therefore, the default emission factor from IFI Dataset is verified as accurate and authoritative.

Based on the above mentioned source, it is verified that the latest $EF_{grid,CM,y}$ of solar project and firm energy project (e.g. Hydro, geothermal) in Cambodia in 2021 is 0.8743 tCO_{2e}/MWh and 0.5878 tCO_{2e}/MWh, respectively.

PP has selected 0.5878 tCO_{2e}/MWh for calculate emission reduction for the project which is verified as conservative.

Furthermore, VVB has searched another data source and assessed as below,

a. IGES List of Grid Emission Factors version 10.12/52/. As per this list, there is only 2009 Phnom Penh grid emission factors of solar project in Cambodia which is 0.6413 tCO_{2e}/MWh. However the data is not

latest and no latest data available as per the list, so this source is confirmed as not applicable for the project.

b. JCM website/53/. By checking the website, it is confirmed that in case the PV system in a proposed project activity in Cambodia is connected to an internal grid but not connected to the national grid, the emission factor can be set to 0.533 tCO₂/MWh as per JCM Approved Methodology KH_AM002. But it is verified that the calculation method of this emission factor is not in line with CDM Tool 07: “Tool to calculate the emission factor for an electricity system” (Version 07.0). So this source is confirmed as not applicable for the project.

Besides, by checking verra website, the emission factor of another registered Cambodia solar project (VCS ID 3613) is 0.7239 tCO₂e/MWh which is higher than 0.5878 tCO₂e/MWh of emission factor for the project.

Therefore, the emission factor of the project is verified as reasonable and conservative.

CL 11 was raised and successfully closed. Refer to Appendix 4 for detail assessment.

3.4.8 Monitoring Plan

1. Data and parameters available at validation

Please refer to the following tables for assessment of each parameter determined ex-ante only used to calculate the ex-ante emission reduction:

<i>EF_{grid,CM,y}</i>	<i>Combined margin CO₂ emission factor for grid connected power generation in year y calculated using TOOL07 (t CO₂/MWh)</i>
Title in line with Methodology?	Yes
Data unit correctly expressed?	Yes
Appropriate description?	Yes
Source clearly referenced? (appropriate?)	IFI Dataset of Default Grid Factors v.3.2 published in April 2022/51/
Correct value provided?	Yes –0.5878 tCO ₂ e/MWh
Has this value been verified?	Confirmed as correct for ex ante determination. Emission factor derived from IFI Dataset of Default Grid Factors v.3.2 published in April 2022/51/ was checked as calculated based on the “Tool to calculate the emission factor for an electricity system”/34/. Via checking all the public source of the Emissions Factor value, it is confirmed that the latest available version that can be obtained before the validation (in line with the requirement of the CDM Tool 07: “Tool to calculate the emission factor for an electricity system” (Version 07.0)) is IFI Dataset of Default Grid Factors v.3.2 which is published in April 2022/51/. So the data source is verified in line with the Tool 07 request.
Choice of data correctly justified?	Yes - Fixed in the first crediting period
Measurement method correctly described?	N/A

2. Data and Parameters Monitored

The monitoring parameters required by the methodology and applicable tools for the project are summarized in the below table.

<i>EG_{facility,y}</i>	<i>Quantity of net electricity generation supplied by the project plant/unit to the grid in year y</i>
Title in line with Methodology?	Yes
Ex-ante Value	Refer to above table 3-5 for ex-ante value and the actual value will be monitored in the monitoring period.
Data unit correctly expressed?	Yes – MWh/yr
Appropriate description?	Yes
Source clearly referenced? (appropriate?)	Yes- EG_{facility,y} will be monitored by PP continuously and recording monthly through the electricity meters as per the PPA/ 11/. EG_{facility,y} is the difference between (a) the quantity of electricity supplied by the project plant to the grid and (b) the quantity of electricity delivered to the project plant from the grid. Via site inspection and checking the PPA/ 11/, it is confirmed that the electricity generation supplied by the project is sent to the 115/22kV Chrak Mtes Substation by the 22kV transmission line, and then delivered to the EDC grid through 115kV transmission line. (a) will be monitored by electricity meters M1 and M2 installed at the H01 STX1 and H03 STX4 of 22kV lines in the low voltage side of the project (b) will be monitored by electricity meters M1, M2 and M3 installed at the H01 STX1, H03 STX4 and H05 HF of 22kV lines in the low voltage side of the project
Correct value provided?	Yes
Has this value been verified?	Yes - The ex-ante values are based on the installed capacity of 26 MWp and annual photovoltaic electricity yield of 1,513.8 kWh/KWp which has been verified as reasonable by checking the photovoltaic electricity yield data from Global Solar Atlas website/48/ by considering the annual degradation rate of solar cell modules of 0.5%
Choice of data correctly justified?	Yes
Measurement method correctly described?	Yes –(a) the quantity of electricity supplied by the project plant to the grid and (b) the quantity of electricity delivered to the project plant from the grid will be monitored by electricity meters and monthly recorded. Designated teams will read and collect the monitored data regularly and recorded in the meter reading records or operation log. The meter readings are also available to the power grid through remote reading system. Monthly settlement for electricity supplied to the grid will be based on the meter readings. Meter reading records will serve as the main data source for emission reductions calculation.
Monitoring frequency correctly described?	Monitored monthly - verified as per the applied methodology
Monitoring equipment correctly described?	N/A
QA/QC procedure correctly described?	Yes - The data will be cross checked with sale and purchase receipts. The electricity meter(s) will be periodically checked and maintained. Calibration will be carried out as per PPA.

	All electronic data and paper documents will be archived for two years following the end of the crediting period.
Purpose of data?	BE calculation

3. Monitoring plan

The VVB has checked the monitoring plan of the Joint-PD-MR against the applied methodology.

The monitoring plan in the Joint-PD-MR has been designed to comply with the latest applicable version of the methodology (ACM0002 Version 21.0).

The validation team evaluated the feasibility and sufficiency of the monitoring plan. The key components of the monitoring plan are as follows.

Monitoring framework:

The Joint-PD-MR contains the Organization Structure of the Monitoring Team to be implemented by the project owner in order to implement the project activity. The VCS monitoring team will be responsible for the monitoring of all the parameters to be monitored including data collection, supervision and witness the whole process of data measuring and recording. The monitoring, recording of electricity meters is to be carried out by designated monitoring staff. In addition, the Project developer appoints internal verifiers who is responsible for internal check of the measurement, collection of relevant sales and purchase receipts, and the calculation of the emission reductions. And all the data will be reviewed by the project developer and VVB. The organizational structure is considered sufficient to fulfil the monitoring requirements of the methodology and to ensure that emission reductions can be verified.

Monitoring parameters, corresponding equipment and installation:

Measurement instruments are described in the monitoring plan as subject to appropriate national standards with respect to installation, accuracy and calibration interval. Main instruments electricity meters are used to monitor related parameter Quantity of net electricity generation supplied by the project plant/unit to the grid $EG_{\text{facility},y}$, which is calculated as difference between (a) the quantity of electricity supplied by the project plant to the grid and (b) the quantity of electricity delivered to the project plant from the grid as assessed in the above table of monitored parameter.

This equipment setup is considered sufficient to carry out the monitoring requirements of the methodology, and the appropriate national standards as listed in JPM for both installation and calibration will be followed.

Quality assurance and quality control procedures:

The Joint-PD-MR contains sufficient description on how quality will be controlled and assured in the monitoring of emission reductions.

Furthermore, error check routines will be established on site and at the point of data storage to detect data measuring/transmission failures as well as malfunctions. If any malfunction of the meters observed, the meter supplier will provide technical support to engage the problem promptly and emission reductions during the corresponding period will be calculated conservatively.

Data management:

All monitoring data will be electronically filed by the end of each month and the electronic data files will be archived in both disk copy and printed hard copy and be kept at least 2 years after the end of the last crediting period.

CAR 11 was raised and successfully closed. Refer to Appendix 4 for detail assessment.

3.5 Non-Permanence Risk Analysis

N/A

4 VERIFICATION FINDINGS

4.1 Accuracy of GHG Emission Reduction and Removal Calculations

1. Assessment of Data and parameters available at validation

Via checking the Joint-PD-MR/1/, it is confirmed that all the ex-ante data and parameters are same to the Joint-PD-MR which is verified as correct. Refer to section 3.4.8 for the detail assessment of ex-ante parameters.

2. Assessment of Data and parameters monitored

During the verification all relevant monitoring parameters (as listed in chapter 3.4.8 of this report) have been verified with regard to the

- (i) appropriateness of the applied measurement / determination method,
- (ii) the correctness of the values applied for ER calculation,
- (iii) the accuracy, and applied QA/QC measures.

The monitoring results as well as the verification procedure are described parameter-wise in the below tables,

<i>EG_{facility,y}</i>	<i>Quantity of net electricity generation supplied by the project plant/unit to the grid in year y</i>
Title in line with Methodology?	Yes
Data unit correctly expressed?	MWh
Appropriate description?	Yes
Monitored Value?	34,694.918
Correct value provided?	Yes
Source clearly referenced? (appropriate?)	Yes – Value is calculated by the difference between (a) the quantity of electricity supplied by the project plant to the grid and (b) the quantity of electricity delivered to the project plant from the grid, both (a) and (b) are derived from meter reading records in Monthly Operation&Maintenance Report/21/
Choice of data correctly justified?	Yes - the measuring and reporting frequency are in line with the monitoring plan and applied methodology/31/.
Measurement method and procedures correctly described?	Yes – (a) the quantity of electricity supplied by the project plant to the grid is measured continuously by bi-directional electricity meters M1 and M2, (a) is calculated by the sum of these two electricity meters readings. (b) the quantity of electricity delivered to the project plant from the grid is measured continuously by electricity meters M1, M2 and M3. (b) is calculated by the sum of these three electricity meters readings. Designated teams read and collect the monitored data monthly and recorded in the meter reading records in Monthly Operation&Maintenance Report/21/. The meter readings are also available to the power grid through remote reading system. Monthly settlement for electricity supplied to the grid are based on the meter readings. Meter reading records served as the main data source for emission reductions calculation for this monitoring period.

	Via checking the measuring method by reviewing the Monthly Operation&Maintenance Report/21/ and interview with monitoring staff, CTI verified that the method is in line with the monitoring plan.																				
Calculation method?	N/A																				
Monitoring frequency correctly described?	Monitoring continuously and recording monthly																				
Monitoring equipment correctly described?	3 electricity meters installed Type: Mk6E For each electricity meter, the calibration information and validity is listed as below table <table><tr><th>Meter No.</th><th>Accuracy</th><th>Serial No.</th><th>Calibrated Date</th><th>Validity</th></tr><tr><td>M1</td><td>0.2S</td><td>920809277</td><td rowspan="3">14-January-2022</td><td rowspan="3">The total metering system accuracy is normal during this monitoring period and no electricity meters need to be re-calibrated.</td></tr><tr><td>M2</td><td>0.2S</td><td>920809279</td></tr><tr><td>M3</td><td>0.5S</td><td>250691873</td></tr></table>					Meter No.	Accuracy	Serial No.	Calibrated Date	Validity	M1	0.2S	920809277	14-January-2022	The total metering system accuracy is normal during this monitoring period and no electricity meters need to be re-calibrated.	M2	0.2S	920809279	M3	0.5S	250691873
Meter No.	Accuracy	Serial No.	Calibrated Date	Validity																	
M1	0.2S	920809277	14-January-2022	The total metering system accuracy is normal during this monitoring period and no electricity meters need to be re-calibrated.																	
M2	0.2S	920809279																			
M3	0.5S	250691873																			
QA/QC procedure correctly described?	Yes – The above calibration information has been confirmed by checking the calibration reports/19/ which has been confirmed by EDC grid as per the requirement from PPA/11/. Via checking the calibration reports/19/, CTI confirmed that all the electricity meters are calibrated based on PPA/11/. And based on Schedule 6 of Metering system, standard and testing in PPA, after installation of such meters, site tests will be undertaken to confirm the accuracy of the meters, if the electricity meters can be used normally, recalibration is not required. Furthermore, as per the Chapter 7.5 of Meter Accuracy in the PPA/11/, the total metering system accuracy shall be 0.2% or better. Should total metering system accuracy be measured above 0.2%, the project owner will immediately report this to the EDC grid for their inspection and arrange for meters to be tested for accuracy and re-calibrated. Via checking the calibration reports/19/, it is confirmed all electricity meters have been calibrated prior to the project start date and accuracy is normal during this monitoring period and no electricity meters need to be re-calibrated. The monitoring data covering this monitoring period have been cross checked with sale and purchase receipts of electricity/22/. Via comparison, the data in the meter reading records in Monthly Operation&Maintenance Report/21/ are verified as consistent with the data in the sale and purchase receipts of electricity/22/ for both (a) and (b). Hence, VVB confirmed that the meter reading records served as the main data source for emission reductions calculation for this monitoring period is reasonable and correct.																				
Purpose of data?	BE calculation																				
Comments?	N/A																				

CAR 12 was raised and successfully closed. Refer to Appendix 4 for detail assessment.

3. Assessment of GHG emission reductions Quantity

Calculation of baseline GHG emissions or removals

According to the methodology and above all the formulars listed in section 3.4.6, baseline emissions are calculated using the following formula:

$$BE_y = EG_{facility,y} \times EF_{grid,CM,y}$$

The values monitored during monitoring survey are transparently shown in the Section 6.1 of Joint-PD-MR. During onsite, CTI cross-checked these values in detail using various supporting records and documents. Refer to the above section 3.4.8 of this report for ex-ante and above section 4.1 for ex-post parameters' assessment.

The Baseline Emissions during this monitoring period is summarized as below,

Date	$EG_{facility,y}$ (MWh)	$EF_{grid,CM,y}$ (tCO _{2e} /MWh)	BE_y (tCO _{2e})
27-January-2022~31-January-2022	311.483	0.5878	183.09
01-February-2022~28-February-2022	3,230.713	0.5878	1,899.01
01-March-2022~31-March-2022	3,288.869	0.5878	1,933.20
01-April-2022~30-April-2022	3,110.759	0.5878	1,828.50
01-May-2022~31-May-2022	3,189.260	0.5878	1,874.65
01-June-2022~30-June-2022	3,333.783	0.5878	1,959.60
01-July-2022~31-July-2022	2,962.886	0.5878	1,741.58
01-August-2022~31-August-2022	3,159.096	0.5878	1,856.92
01-September-2022~30-September-2022	2,922.771	0.5878	1,718.00
01-October-2022~31-October-2022	3,014.458	0.5878	1,771.90
01-November-2022~30-November-2022	2,753.017	0.5878	1,618.22
01-December-2022~31-December-2022	3,417.823	0.5878	2,009.00
Total	34,694.918	0.5878	20,393

In conclusion, the calculations of baseline GHG emissions have been carried out in accordance with the formulae and methods described in the registered monitoring plan, the applied methodology and, where applicable, the applied standardized baseline. Any assumptions used in emission calculations have been justified. Appropriate emission factors and other reference values have been correctly applied. No errors, miscalculations, omissions, misstatements or incomplete information has been identified. It can be confirmed that the baseline calculation is overall correct.

Calculation of project GHG emissions or actual net GHG removals by sinks

According to the methodology, the project emissions $PE_y = 0$.

Calculation of leakage GHG emissions

According to the methodology, the leakage emissions are not considered.

Summary of calculation of GHG emission reductions or net anthropogenic GHG removals by sinks

The VVB has checked the Joint-PD-MR includes a summary table of the emission reductions calculation.

Summary of emission reductions during the monitoring period,

$$ER_y = BE_y - PE_y = BE_y$$

Hence, the emission reductions during this monitoring period are summarized in the table below

Parameters Period	Baseline Emissions BE_y	Project Emissions PE_y	Net GHG emission reductions or removals ER
	(tCO ₂ e)	(tCO ₂ e)	(tCO ₂ e)
27-January-2022~31-December-2022	20,393	0	20,393
Total	20,393	0	20,393

All the figures as per the monitoring report were cross-checked by the VVB against basic monitored data. Refer to section 4.4.1 and 4.4.2 of this report for assessment details.

4.2 Quality of Evidence to Determine GHG Emission Reductions and Removals

Refer to section 4.1 above for the detail assessment of each monitoring parameters and Appendix 3 for the supporting evidence used to determine the GHG emission reductions.

For each reported data, the evidence is provided and verified as sufficient and quality is appropriate. Also, the cross-checks have been performed on the reported data with different source of evidence. The information flow from data generation and aggregation, to recording, calculation and final transposition into the monitoring report has been assessed by VVB in section 4.1 for each parameter and also the calibration have been conducted as per the frequency of monitoring equipment defined in the JPM.

Therefore, it is concluded that the evidence provided are verified as sufficient and quality is appropriate and thus the evidence can be used to determine the GHG emission reductions and removals for this monitoring period.

5 VALIDATION AND VERIFICATION OPINION

Climate Bridge (Shanghai) Ltd. has commissioned the CTI to carry out the Verified Carbon Standard (VCS) joint validation and 1st periodic verification of the project activity 26MWp Solar Photovoltaic Power Plant in Bavet City, Cambodia (VCS ID 4036), for the monitoring period from 27-January-2022 to 31-December-2022, with regard to the relevant requirements for VCS Standard Version 4.4.

Based on the reviewing the documented evidence and by an on-site assessment, CTI can confirm that:

- the baseline scenario is correctly defined as per the applied methodology and relate tools;
- the project is additional;
- all data and information used for ex-ante calculation of emission reductions is of projected and/or hypothetical nature;
- the monitoring plan in the validated Joint-PD-MR is as per the applied baseline and monitoring methodology.
- the project has been implemented and operated as per the VCS Joint-PD-MR;
- the project complies with the validation and verification criteria for projects set out in VCS Standard Version 4.4;
- the monitoring report and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS requirement and ACM0002 (version 21.0);
- the monitoring is in place as per the applied baseline and monitoring methodology ACM0002 (version 21.0);
- the monitoring complies with the monitoring plan in the validated PD;

The estimated GHG emission reductions from the project during the crediting period amount as follows:

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
27-January-2022~31-December-2022	21,487
01-January-2023~31-December-2023	23,027
01-January-2024~31-December-2024	22,912
01-January-2025~31-December-2025	22,797
01-January-2026~31-December-2026	22,683
01-January-2027~31-December-2027	22,570
01-January-2028~31-December-2028	22,457
01-January-2029~26-January-2029	1,599
Total estimated ERs	159,532
Total number of crediting years	7

Average annual ERs	22,790
--------------------	--------

It is certified that the GHG emission reductions from the project during the monitoring period amount as follows:

Verification period: From 27-January-2022 to 31-December-2022

Verified GHG emission reductions and removals in the above verification period by vintage is

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
27-January-2022~31-December-2022	20,393	0	0	20,393
Total	20,393	0	0	20,393

Year	Ex-ante emissions reductions/removals	Achieved emissions reductions/removals	Percent difference	Justification for the difference
27-January-2022~31-December-2022	21,487	20,393	-5.09%	The mainly difference between the ex-ante estimated value and actual value is due to normal fluctuation of solar energy and is deemed reasonable.
Total	21,487	20,393	-5.09%	

Shunrong Lin

Lin Shunrong

CTI

Verification Team Leader

18-December-2023

Wu Lin

Lin Wu

CTI

Technical Reviewer

18-December-2023

APPENDIX 1: ABBREVIATIONS

Abbreviations	Full texts
BE	Baseline Emission
BESS	Battery Energy Storage System
BM	Build Margin
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CER	Certified Emission Reduction
CL	Clarification Request
CM	Combined Margin
CO ₂	Carbon dioxide
CP	Crediting Period
DNA	Designated National Authority
EB	Executive Board
EDC	Electricite Du Cambodge
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reduction
FAR	Forward Action Request
GHG	Green House Gas
GS	Gold Standard
GSC/GSP	Global Stakeholder Consultation Process
IPCC	Intergovernmental Panel on Climate Change
IRR	Internal Rate of Return
JPM	Joint-PD-MR
KP	Kyoto Protocol
LE	Leakage Emission
LSC	Local Stakeholder Consultation
MME	Ministry of Mines and Energy
MoV	Means of Validation
MP	Monitoring Plan
NBC	National Bank of Cambodia
OM	Operating Margin
PA	Project Activity
PE	Project Emission
PP	Project Participant
PS	Project Standard
QC/QA	Quality control/Quality assurance
SD	Sustainable Development
SDG	Sustainable Development Goals
tCO ₂ e	Tonnes of Carbon di oxide equivalent
UNFCCC	United Nations Framework Convention on Climate Change
VVS	Validation and Verification Standard
VCS	Verified Carbon Standard
VCSA	Verified Carbon Standard Association
VCU	Verified Carbon Unit
VVB	Validation/Verification Body

APPENDIX 2: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS

Ms. Shunrong LIN

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification						
Status	GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
Date	√	√	√	√	√	√

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources
SS 3: Energy demand	TA 3.1: Energy demand
SS 13: Waste handling and disposal	TA 13.1: Solid waste and wastewater
	TA 13.2: Manure
SS 14: Afforestation and reforestation	TA 14.1: Afforestation and reforestation
SS 15: Agriculture	TA 15.1: Agriculture

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:

Wu Lin

Wu LIN

Technical Competent Manager

Shenzhen, 25/12/2022

Mr. Emanuel Rogier van Mansvelt

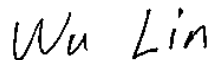
Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification
Local Expert

Appointed Geography Area As Local Expert
Cambodia

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:



Wu LIN

Technical Competent Manager

Shenzhen, 05/10/2022

Mr. Wu LIN

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification						
Status	GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
Date	✓	✓	✓	✓	✓	✓

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.1: Thermal energy generation
	TA 1.2: Energy generation from renewable energy sources
SS 2: Energy distribution	TA 2.1: Electricity distribution
SS 3: Energy demand	TA 3.1: Energy demand
SS 4: Manufacturing industries	TA 4.1: Cement and lime production
SS 5: Chemical industry	TA 5.1: Chemical industry
	TA 5.2: Caprolactam, nitric and adipic acid
SS 10: Fugitive emissions from fuels (solid, oil and gas)	TA 10.1: Fugitive emissions from oil and gas
SS 11: Fugitive emissions from production and consumption of halocarbons and sulphur hexafluoride	TA 11.1: Emissions of fluorinated gases
	TA 11.2: Refrigerant gas production
SS 12: Solvents use	TA 12.1: Chemical industry
SS 13: Waste handling and disposal	TA 13.1: Solid waste and wastewater
	TA 13.2: Manure

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:

Lu ZHOU



General Manager

Shenzhen, 01/01/2021

APPENDIX 3: DOCUMENTS REVIEWED OR REFERENCED

No.	Author	Title	References to the document	Provider
1.	PP	Joint-PD-MR of “26MWp Solar Photovoltaic Power Plant in Bavet City, Cambodia”	- Draft Version No. 1.1, dated 20-February-2023 - Final Version No. 1.2, dated 26-June-2023 - Final Version No. 1.3, dated 06-December-2023	PP
2.	PP	Emission Reduction Calculation spreadsheet of “26MWp Solar Photovoltaic Power Plant in Bavet City, Cambodia” – Ex-ante calculation combined with 1 st monitoring period	- Draft Version No. 1.0, dated 20-February-2023 - Final Version No. 1.2, dated 26-June-2023 - Final Version No. 1.3, dated 06-December-2023	PP
3.	PP	IRR calculation sheet	- Draft Version No. 01, dated 20-February-2023 - Final Version No. 02, dated 26-June-2023 - Final Version No. 03, dated 06-December-2023	PP
4.	Minister of Commerce	Business License of PP	Business License of Green Sustainable Ventures (Cambodia) Co., Ltd.	PP
5.	A.B.Z Group Co.,Ltd.	Predesign Studies	Completed in October 2019	PP
6.	Ministry of Mines and Energy	Project approval	Project approval issued on 12-July-2019	PP
7.	aNew Energy	Financial investment analysis sheet	Financial investment analysis sheet	PP
8.	PP	Investment details	Investment details sheet & consulting fee	PP
9.	ERD Consultants Co., Ltd.	Environment Impact Assessment (EIA)	Issued in July 2021	PP
10.	Ministry of Environment	EIA approval	Issued on 30-November-2021	PP
11.	Electricite Du Cambodge (EDC) Grid with PP	Power Purchase Agreement (PPA)	1. issued on 27-August-2019 2. issued on 13-October-2020 3. issued on 23-August-2021	PP
12.	PP and Petech (Cambodia) PLC.	Engineering Procurement Construction (EPC) contract	Signed on 13-January-2020	PP

13.	Ministry of Land Management, Urban Planning and Construction in Cambodia	Construction Permit	Issued on 13-April-2021	PP
14.	Manufacturers	Technical agreement	Technical agreement of main equipment solar cell modules and inverters	PP
15.	VVB	Nameplate of the equipment	Nameplate of the equipment, checked by VVB during site visit	VVB
16.	Land owner with PP	Lease Agreement	Issued on 22-January-2020	PP
17.	EDC grid	Commissioning Letter	Issued on 27-January-2022	PP
18.	PP	Technical flow chart	Technical flow charts in the project site	PP
19.	EDC grid	Calibration Reports	Calibration Reports to all the electricity meters issued on 14-January-2022	PP
20.	PP	Operation log	Operation log of the project	PP
21.	PP	Monthly Operation & Maintenance Report	Monthly Operation & Maintenance Report issued by PP for this monitoring period	PP
22.	PP and EDC grid	Electricity sales & purchase receipts	Electricity sales & purchase receipts for this monitoring period	PP
23.	PP	Health Safety Environment monthly report	Health Safety Environment monthly report	PP
24.	PP	Local Stakeholder Consultation Records during 1 st monitoring period	Records of the survey during the 1 st monitoring period	PP
25.	PP	Declaration of no double counting and not involved in other GHG scheme	Issued on 19-June-2023	PP
26.	PP and employees	Staff List	Staff List signed with employees for implementation of this project	PP
27.	United Nations	Least Developed Country definition	https://www.un.org/en/conferences/least-developed-countries	Public website
28.	PP	Report	Report of Cambodia Socio-Economic Survey 2021	Public Website
29.	International Finance Corporation	Guidelines	International Finance Corporation Environmental, Health, and Safety Guidelines https://www.ifc.org/content/dam/ifc/doc/2023/ifc-general-ehs-guidelines.pdf	Public Website
30.	National Bank of Cambodia	Local lending rate	NBC Annual Report 2018 issued on 30-Jan-2019, https://www.nbc.org.kh/english/index.php	Public Website

31.	UNFCCC	CDM Approved Methodology ACM0002	“Grid-connected electricity generation from renewable sources” (version 21.0)	UNFCCC website
32.	UNFCCC	Methodological tool	Tool for the demonstration and assessment of additionality (Version 07.0)	UNFCCC website
33.	UNFCCC	Methodological tool	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (Version 3.0)	UNFCCC website
34.	UNFCCC	Methodological tool	Tool to calculate the emission factor for an electricity system (version 07.0)	UNFCCC website
35.	UNFCCC	Methodological tool	Common practice (Version 03.1)	UNFCCC website
36.	UNFCCC	Methodological tool	Investment analysis (version 12.0)	UNFCCC website
37.	VERRA	Template	VCS-Joint-Project-Description-Monitoring-Report-Template-v4.2	VCS website
38.	VERRA	Registration and Issuance Process	Registration and Issuance Process version 4.3	VCS website
39.	VERRA	VCS Program Definitions	VCS Program Definitions, v4.3	VCS website
40.	VERRA	VCS Program Guide	VCS Program Guide, v4.3	VCS website
41.	VERRA	VCS Standard	VCS Standard Version 4.4	VCS website
42.	UNFCCC	UNFCCC website	https://cdm.unfccc.int	UNFCCC website
43.	VERRA	VCS	https://verra.org/project/vcs-program/	VCS website
44.	GS	GS	http://globalgoals.goldstandard.org/	GS website
45.	VVB	Site Visit Photo	Photo taken by VVB during site visit including main equipment, monitoring devices, etc.	N/A
46.	National Government	Law&Regulation	1. The Law on Environmental Protection and Natural Resource Management 2. Electricity Law of the Kingdom of Cambodia 3. Land Management rule 4. Long-term strategy for Carbon Neutrality 5. Cambodia Basic Energy Plan 6. Law on Environmental Protection and Natural Resource Management	Public Website
47.	PWC	Cambodian 2018 Tax Booklet	https://www.pwc.com/kh/en/tax/cambodian-2018-tax-booklet1.pdf	Public Website
48.	Global Solar Atlas	Global Solar Atlas website	https://globalsolaratlas.info/map?c=11.119117,105.837922,10&s=11.088123,106.066388&m=site	Public Website

49.	EDC grid	EDC grid website	https://www.edc.com.kh/Annually_page/annuallyReport	Public Website
50.	Ministry of Environment of Cambodia	Designated National Authority (DNA) in Cambodia	https://ncsd.moe.gov.kh/dcc/program/cambodia-climate-change-alliance-ccca-phase-i#:~:text=The%20Ministry%20of%20Environment%20%28MoE%29%20is%20the%20Focal,Authority%20%28DNA%29%20for%20the%20Clean%20Development%20Mechanism%20%28CDM%29.https://www.moe.gov.kh/	Public Website
51.	UNFCCC	IFI Dataset of Default Grid Factors v.3.2 published in April 2022	https://unfccc.int/climate-action/sectoral-engagement/ifi-harmonization-of-standards-for-ghg-accounting/ifi-twg-list-of-methodologies	Public Website
52.	IGES	IGES List of Grid Emission Factors version 10.12	https://www.iges.or.jp/en/pub/list-grid-emission-factor/en	Public Website
53.	JCM	JCM website	https://www.jcm.go.jp/kh-jp/methodologies/46/approved_pdf_file	Public Website
54.	PP	Certified Articles of Incorporation and Attested Ownership Structure	Certified Articles of Incorporation and Attested Ownership Structure of the project owner	PP
55.	PP and PESTECH (Cambodia) Company	O&M Contract	GSV PCL O&M Contract (Executed) dated 29-January-2022	PP
56.	Average Salary Survey	Public data of the average salary level in Cambodia	https://www.averagesalarysurvey.com/zh/salary/cambodia	Public Website
57.	PP	Insurance list and related Insurance Policy	Insurance list including all the Insurance bought and related fees Related Insurance Policy	PP

APPENDIX 4: CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS AND FORWARD ACTION REQUESTS

Table 1.Remaining FAR from validation and/or previous verification

FAR ID	N/A	Section no.		Date:
Description of FAR				
Project participant response				Date:
Documentation provided by project participant				
VVB assessment				Date:

Table 2.CL from this verification

CL ID	01	Section no.	1.3	Date : 17/04/2023
Description of CL				
In section 1.3, 1. PP stated that “project does not belong to projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal, or destruction”, however, the reason is not clarified clearly. 2. PP does not clarify how the project is in line with the requests from section 3.1 of the VCS standard version 4.4.				
Project participant response				Date: 30/05/2023
1. The project utilizes the solar energy to generate and supply electricity to EDC grid which is dominated by coal-fired power plants. Baseline emissions of the project include only CO ₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The project assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants, which is in line with the methodology ACM0002 (Version 21.0). Therefore, the baseline emission cannot reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal, or destruction. 2. The clarification has been added in the updated JPM.				
Documentation provided by project participant				
/1/- version 2.0				
VVB assessment				Date: 16/06/2023
1. The revised Joint-PD-MR is checked, CTI confirmed that the clarification has been added. 2. The revised Joint-PD-MR is checked, CTI confirmed that the clarification has been added. Refer to section 3.1 of this report for the detail assessment. CL 01 is closed.				

CL ID	02	Section no.	1.7	Date : 17/04/2023
Description of CL				
PP did not clarify the clear ownership in section 1.7 as per the information given in the section 2.2 of JPM, especially for the land ownership of the project site.				
Project participant response				Date : 30/05/2023
The relevant information especially for land ownership has been added in the section 1.7 of the updated JPM.				
Documentation provided by project participant				

/1/- version 2.0 /10/ /6/ /12/ /4/ /16/ /11/	
VVB assessment	Date: 16/06/2023
The section 1.7 of the Joint-PD-MR has been checked, it is confirmed that the Project Proponent demonstrates that they have the legal right to control and operate project activity. Via checking the National legal approval to the Environmental Impact Assessment (EIA)/10/, Project Approval/6/, EPC contract/12/, Power Purchase Agreement (PPA)/11/ and the business license of the Green Sustainable Ventures (Cambodia) Co., Ltd./4/, it is verified that the project owner of this solar project is Green Sustainable Ventures (Cambodia) Co., Ltd. and it has the legal right to control and operate the project activity. By checking Certified Articles of Incorporation and Attested Ownership Structure of the project owner/54/, it is verified that Pestech owned 100% shares of Astoria Solar Farm SDN.BHD who owned 94% equity of project owner. Besides, Salah Essa, a Cambodian individual owned 6% of the equity of project owner. As stated by project owner, aNew Energy, an energy investment expert, introduced Astoria Solar Farm SDN.BHD which is wholly owned by Pestech to establish the project owner company with Salah Essa together. Furthermore, via checking the Land Lease Agreement/16/ of the project site, it is confirmed that Salah Essa (land owner) who owns the project land has leased the project land to the project owner for 20 years to operate this solar project indicating that the project owner has the land use right to the project land for 20 years. Thus, it is confirmed that the project owner has the ownership of the project plant, equipment and power generating and carbon credits generated from this project. CL 02 is closed.	

CL ID	03	Section no.	1.14	Date : 17/04/2023
Description of CL				
How the project in compliance with each listed Laws, Statutes and Other Regulatory Frameworks is not clarified respectively.				
Project participant response				Date : 30/05/2023
The relevant clarification has been added in the section 1.14 of the updated JPM.				
Documentation provided by project participant				
/1/- version 2.0 /10/ /6/ /46/				
VVB assessment				Date: 16/06/2023
The revised Joint-PD-MR is checked, CTI confirmed that the clarification has been added. The Joint-PD-MR has stated that the project is in the renewable energy power generation industry, this industry is confirmed as in line with the laws and regulations/46/ as listed in the Joint-PD-MR. Also the project is legally approved by Local government by checking the EIA approval/10/ and Project Approval/6/, which means that the project is in line with the national and local laws and regulations based on the local expertise of the local expert in the audit team. And as stated in the Joint-PD-MR/1/ and confirmed by interview with local officers/12/, CTI verified that the project will be subject to regular inspection by local government during the implementation period to ensure continuous compliance. In conclusion, the project complies with all relevant local, regional and national laws, statutes and regulatory frameworks in Cambodia. CL 03 is closed.				

CL ID	04	Section no.	2.2	Date : 17/04/2023
Description of CL				
In section 2.2, 1. The stakeholder is not identified for the project activity. 2. The actual local stakeholder consultation process prior to the project construction is not clarified.				
Project participant response				Date : 30/05/2023
1. The stakeholder has been identified in the section 2.2 of the updated JPM. 2. The relevant clarification has been added in the section 2.2 of the updated JPM.				
Documentation provided by project participant				
/1/- version 2.0				
VVB assessment				Date: 16/06/2023
1. The revised Joint-PD-MR is checked, CTI confirmed that the actual local stakeholder has been identified which are verified as relevant. 2. The revised Joint-PD-MR is checked, CTI confirmed that the actual local stakeholder consultation process prior to the project construction is clarified in detail. Refer to section 3.3.2 of this report for detail assessment. CL 04 is closed.				

CL ID	05	Section no.	2.2	Date : 17/04/2023
Description of CL				
In section 2.2, for Local Stakeholder Consultation during the project implementation stage, 1. How the interviews were conducted by PP in February 2022 with local stakeholders is not clarified. 2. How many people interviewed in February 2022 is not clarified. 3. PP stated that in March 2022, the project received a comment, however, how to obtain this comment is not clarified. 4. The evidence of this comment and how to solve this comment is not provided and clarified. 5. PP listed a picture of contact method, however, it is not clarified what is used for.				
Project participant response				Date : 30/05/2023
In section 2.2, for Local Stakeholder Consultation during the project implementation stage, 1. In February 2022, the project owner visited 2 of land sellers, 1 of village chief, 4 of villagers and 6 of the project workers face to face to collect their comments, all of them expressed satisfaction with the project. 2. 13 people were interviewed in February 2022. 3. In March 2022, the maintenance team of the project went to trim the trees to avoid tripping over the power transmission line, but the nearby Pagoda Monks directly prevented the team from trimming. 4. Since the first item of the grievance redress mechanism could not solve the problem, the project manager invited the village chief to discuss with the Monk head and explained that the tree branches that were too high would trip the transmission line and cause problems such as electric shock. After mediation, the Monk head understood the importance of pruning the branches and agreed to do so. Monthly operation & maintenance has been provided. 5. A grievance signage shown in Figure 2-4 has been placed out the project gate for anyone to provide feedback. The relevant response has been added in the updated JPM.				
Documentation provided by project participant				
/1/- version 2.0				
VVB assessment				Date: 16/06/2023
1. The revised Joint-PD-MR is checked, CTI confirmed that the requested information has been clarified. 2. The revised Joint-PD-MR is checked, CTI confirmed that the number of people has been clarified. 3. The revised Joint-PD-MR is checked, CTI confirmed that the requested information has been clarified. 4. The revised Joint-PD-MR is checked, CTI confirmed that the requested information has been clarified.				

5. The revised Joint-PD-MR is checked, CTI confirmed that the requested information has been clarified.
Refer to section 3.3.2 of this report for detail assessment of above issues.
CL 05 is closed.

CL ID	06	Section no.	2.3	Date : 17/04/2023
Description of CL				
In section 2.3, 1. If the environmental impacts analysis in line with the related Law on Environmental Protection and Natural Resource Management is not clarified. 2. PP stated that “The procedures of land acquisition are in compliance with the relevant laws in Cambodia”, however, the law is not clarified in detail. 3. PP stated that “Noise level measurements has been conducted to ensure that noise level is complied with relevant national standard”, however, the national standard is not clarified in detail.				
Project participant response				Date : 30/05/2023
In section 2.3, 1. The final EIA report of the project was approved by Ministry of Environment on 30-Nov-2021, indicating that EIA is in compliance with The Law on Environmental Protection and Natural Resource Management. 2. Prior to construction, the project has acquired the construction permit issued by Ministry of Land Management on 13-Apr-2021. The procedures of land acquisition are in compliance with the Land Management in Cambodia. 3. During the operation period, the main noise is generated by inverters and transformers. Operation of noise-generating equipment are enclosed, and periodic maintenance of equipment will be conducted. Noise level measurements has been conducted to ensure that noise level is complied with International Finance Corporation Environmental, Health, and Safety Guidelines. The relevant response has been added in the updated JPM.				
Documentation provided by project participant				
/1/- version 2.0 /9/ /10/ /46/ /29/				
VVB assessment				Date: 16/06/2023
1. The revised Joint-PD-MR is checked, CTI confirmed that related information has been added. The environmental impacts of the project were sufficiently assessed by means of an Environmental Impact Assessment (EIA) according to the “Law on Environmental Protection and Natural Resource Management”/46/. Every aspect of environmental impact has been considered in the EIA report with corresponding measures during project construction and operation. 2. The revised Joint-PD-MR is checked, CTI confirmed that related information has been clarified, via checking the construction permit issued by Ministry of Land Management on 13-April-2021/13/, it is confirmed that procedures of land acquisition are in compliance with the Land Management in Cambodia. 3. The revised Joint-PD-MR is checked, CTI confirmed that related information has been clarified. There were no exhaust gas emissions, no domestic wastewater and solid waste observed by site inspection but noise heard which is generated by inverters and transformers, however, noise level measurements has been conducted to ensure that noise level is complied with International Finance Corporation Environmental, Health, and Safety Guidelines/29/. And by site inspection, it is confirmed that the lowest distance from project site to villager’s home is more than 1km, and as confirmed by the local resident who lived in that village, it is verified that no noise contribution to the surrounding boundary after distance attenuation. Hence VVB verified that noise has little influence on the boundary acoustic environment which will not impact the local residents. CL 06 is closed.				

CL ID	07	Section no.	3.2	Date : 17/04/2023
Description of CL				
In section 2.3, PP has justified the applicability conditions of the Tool to calculate the emission factor for an electricity system (Version 07.0), but PP stated that “Based on the public data in Cambodia, either the specific fuel consumption of the EDC grid system in most recent five years or CO ₂ emission factor of each plant are not available, lead to the combined margin emission factor (EF _{grid,CM,y}) cannot be calculated by this tool”, then why this tool is applicable to this project is not clarified.				
Project participant response				Date : 30/05/2023
Based on the public data in Cambodia, either the specific fuel consumption of the EDC grid system in most recent five years or CO ₂ emission factor of each plant are not available, lead to the combined margin emission factor (EF _{grid,CM,y}) cannot be calculated by the latest version “Tool to calculate the emission factor for an electricity system”. Besides, there is no available latest emission factor of the EDC grid as approved by the relevant National Authority or Designated National Authority (DNA) in Cambodia. By checking the UNFCCC website, for the purpose of promoting greater harmonization, the IFI Technical Working Group (IFI TWG) on GHG accounting maintains a common dataset containing Default Emissions Factor (DEF) of the country’s electricity grid including in-country interconnected grids. The DEFs apply to electricity generation in a country and currently do not consider the impact of interconnections with neighbouring countries. According to the public “Methodological Approach for the Common Default Grid Emission Factor Dataset”, the common dataset containing DEFs is constructed using a Combined Margin (CM) for the grid that is calculated by combining an Operating Margin (OM) and a Build Margin (BM) as per CDM Tool 07: “Tool to calculate the emission factor for an electricity system” (Version 07.0). This calculation method of CM is in line with ACM0002 (Version 21.0). Besides, the country specific information that is used to calculate OM and BM is from The International Energy Agency’s (IEA) energy statistics database. Therefore, the default emission factor from IFI Dataset is accurate and authoritative. According to the latest IFI Dataset of Default Grid Factors v.3.2 published in April 2022, the latest EF _{grid,CM,y} of solar project and firm energy project (e.g. <i>Hydro, geothermal</i>) in Cambodia in 2021 is 0.8743 tCO ₂ e/MWh and 0.5878 tCO ₂ e/MWh, respectively. To be conservative, 0.5878 tCO ₂ e/MWh can be used to calculate emission reduction for the project.				
Documentation provided by project participant				
/1/- version 2.0 /51/				
VVB assessment				Date: 16/06/2023
The revised Joint-PD-MR is checked, CTI confirmed that related information has been added. The conservative value of Default Emissions Factor (DEF) issued by IFI Technical Working Group (IFI TWG)/51/ has been used as the emission factor for the grid of the project. Via checking all the public source of the Emissions Factor value, it is confirmed that the latest available version that can be obtained before the validation (in line with the requirement of the CDM Tool 07: “Tool to calculate the emission factor for an electricity system” (Version 07.0)) is IFI Dataset of Default Grid Factors v.3.2 which is published in April 2022/51/. So the data source is verified in line with the Tool 07 request. Hence this criteria is applicable. CL 07 is closed.				

CL ID	08	Section no.	3.5	Date : 17/04/2023
Description of CL				
For the additionality demonstration, the project shall demonstrate regulatory surplus at validation as per 3.14.1 of VCS standard. Clarification is requested.				
Project participant response				Date : 30/05/2023
As per the relevant applicable laws and regulations at validation, i.e. Electricity Law of The Kingdom of				

Cambodia, Long-term strategy for Carbon Neutrality, Cambodia Basic Energy Plan, the project activity is in compliance with all laws and regulations in Cambodia and the project activity is not mandated by any law, statute, or other and regulations. The same has been provided in the updated JPM.	
Documentation provided by project participant	
/1/- version 2.0	
VVB assessment	Date: 16/06/2023
The revised Joint-PD-MR is checked, the clarification has been added. Via searching the public website with laws and regulations in Cambodia, it is confirmed that there is no legal law and regulation to mandate to implement solar energy for electricity generation in Cambodia by checking the Electricity Law of The Kingdom of Cambodia, Long-term strategy for Carbon Neutrality, Cambodia Basic Energy Plan/46/. Furthermore, via site interview with local officers, it is also confirmed that using renewable energy for electricity generation are encouraged by the state and not mandatory. Therefore, the project activity is not mandated by any law, statute, or other regulatory framework, it is deemed as regulatory surplus. CL 08 is closed.	

CL ID	09	Section no.	3.5	Date : 17/04/2023
Description of CL				
In section 3.5, for Sub-step 1b: Consistency with mandatory laws and regulations , PP is not clarified how the alternatives (a) and (b) are consistent with mandatory laws and regulations.				
Project participant response				Date : 30/05/2023
According to Cambodia Basic Energy Plan prepared by The General Department of Energy, Ministry of Mines and Energy of Cambodia (MME), the MME will open the power generation market to local and foreign entities such as the Asian Development Bank and private companies to install solar/PV in Cambodia for supplying electricity to local people. There is no legal prohibition regarding the implementation of the project currently or in the future, therefore alternative (a) is in compliance with mandatory laws and regulations. Alternative (b) Continuation of the current situation is that electricity delivered to EDC grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources. As per the Electricity Law of The Kingdom of Cambodia, there are no laws or regulations in Cambodia that prohibit the power grid from providing the same annual electricity generation. Therefore alternative (b) is in compliance with mandatory laws and regulations. The same has been provided in the updated JPM.				
Documentation provided by project participant				
/1/- version 2.0				
VVB assessment				Date: 16/06/2023
The revised Joint-PD-MR is checked, the clarification has been added. Via searching the public website with laws and regulations in Cambodia, it is confirmed that there is no legal prohibition regarding the implementation of the project currently or in the future by checking the Cambodia Basic Energy Plan/46/, and continuation of the current situation is that electricity delivered to EDC grid is still in line with the Electricity Law of The Kingdom of Cambodia/46/ and no laws or regulations in Cambodia that prohibit the power grid from providing the same annual electricity generation. As assessed above, CTI confirmed that the above alternatives (a) and (b) are considered to follow all mandatory applicable legal and regulatory requirements which are verified based in validation team's local expertise. CL 09 is closed.				

CL ID	10	Section no.	3.5	Date : 17/04/2023
Description of CL				
In section 3.5, 1. Why the action of project owner sent the letter for the project to Ministry of Mines and Energy (MME)				

on 10-May-2019 and approved by MME on 12-Jul-2019 is identified as investment decision is not clarified.

2. The demonstration of prior consideration of carbon revenue is not clarified.

3. For Electricity Tariff, the requirement from section 34 of the applied methodology need to be clarified.

4. Why the input values of financial parameters derived from financial investment analysis sheet prior to the construction of the project is reasonable is not clarified.

5. It is observed that the IRR value with carbon revenue is still lower than benchmark and not changed much comparing with the IRR without carbon revenue, why PP invest the project by considering carbon revenue is not clarified.

Project participant response

Date : 30/05/2023

In section 3.5,

1. Since the PPA was signed by both project owner and the EDC grid on 27-August-2019, this date is the earliest date that can record investment actions.

2. The super majority of project owner - Pestech was first introduced by aNew Energy to the sole owner of project owner back in 2019. aNew Energy originates and co-develops projects related to wind, solar and energy storage with their partners and their clients. aNew Energy also assists in merger and acquisition, fund raise, and market connect in Asia for energy related companies. aNew Energy specializes in structuring projects with their partners. aNew Energy strives to match investors risk-reward profiles to the relevant projects. Financial investment analysis sheet prior to the construction was finished by Camillus Yang of aNew Energy in May 2019. Camillus has a passion for integrated renewable energy projects and matchmaking complimenting investors and technology to projects. He has most renewable energy experience and enjoys developing an options pool (structuring, financing, technology) for clients and partners alike. Therefore, the professionalism and accuracy of this financial investment analysis sheet can be guaranteed. Moreover, the completion date of this financial investment analysis sheet is very close to the investment decision date, indicating this financial investment analysis sheet is reasonable as an investment basis. The project owner made an investment decision on 27-August-2019 to develop the project as a GHG emission reduction project such as VCS project based on this financial investment analysis sheet. This investment decision was made before the starting date of the project, indicating that VCS was seriously considered in this decision prior to the implement of the project.

3. As per the Power Purchase Agreement (PPA) signed by the project owner and the EDC grid, electricity tariff prior to the commercial date is 42 USD/MWh, 76 USD/MWh of electricity tariff is fixed until the end of the term of the PPA and the highest tariff that they received by supplying the electricity to the EDC grid.

4. please refer to 2 above for details.

5. It is calculated that the pre-tax project IRR without the carbon revenues is 7.88% which is lower than the 10.61% of benchmark rate. This demonstrates that the proposed project activity is not a commercially viable option to supply power. Though the pre-tax project IRR with the carbon revenues is 9.09% which is still lower than the benchmark, the return on investment for the project with carbon revenues can be improved to a certain extent. Therefore, the pre-tax IRR with carbon revenues of 9.09% is acceptable to the project owner.

The same above has been provided in the updated JPM.

Documentation provided by project participant

/1/- version 2.0

VVB assessment

Date: 16/06/2023

1. The revised Joint-PD-MR is checked, CTI confirmed that related information has been added.

2. The revised Joint-PD-MR is checked, CTI confirmed that related information has been added.

3. The revised Joint-PD-MR is checked, CTI confirmed that related information has been added.

4. The revised Joint-PD-MR is checked, CTI confirmed that related information has been added.

5. The revised Joint-PD-MR is checked, CTI confirmed that related information has been added.

Refer to section of 3.4.5 of this report for detail assessment of above issues.

CL 10 is closed.

CL ID	11	Section no.	3.6	Date : 17/04/2023
Description of CL				
For final decision of the grid emission factor $EF_{grid,CM,y}$, why the most conservative value from the data sources is not used as the final emission factor is not clarified.				
Project participant response				Date : 30/05/2023
Based on the public data in Cambodia, either the specific fuel consumption of the EDC grid system in most recent five years or CO₂ emission factor of each plant are not available, lead to the combined margin emission factor ($EF_{grid,CM,y}$) cannot be calculated by the latest version “Tool to calculate the emission factor for an electricity system”. Besides, there is no available latest emission factor of the EDC grid as approved by the relevant National Authority or Designated National Authority (DNA) in Cambodia. By checking the UNFCCC website, for the purpose of promoting greater harmonization, the IFI Technical Working Group (IFI TWG) on GHG accounting maintains a common dataset containing Default Emissions Factor (DEF) of the country’s electricity grid including in-country interconnected grids. The DEFs apply to electricity generation in a country and currently do not consider the impact of interconnections with neighbouring countries. According to the public “Methodological Approach for the Common Default Grid Emission Factor Dataset”, the common dataset containing DEFs is constructed using a Combined Margin (CM) for the grid that is calculated by combining an Operating Margin (OM) and a Build Margin (BM) as per CDM Tool 07: “Tool to calculate the emission factor for an electricity system” (Version 07.0). This calculation method of CM is in line with ACM0002 (Version 21.0). Besides, the country specific information that is used to calculate OM and BM is from The International Energy Agency’s (IEA) energy statistics database. Therefore, the default emission factor from IFI Dataset is accurate and authoritative. According to the latest IFI Dataset of Default Grid Factors v.3.2 published in April 2022, the latest $EF_{grid,CM,y}$ of solar project and firm energy project (e.g. <i>Hydro, geothermal</i>) in Cambodia in 2021 is 0.8743 tCO₂e/MWh and 0.5878 tCO₂e/MWh, respectively. To be conservative, 0.5878 tCO₂e/MWh can be used to calculate emission reduction for the project. Besides, by checking verra website, the emission factor of another Cambodia solar project (VCS ID 3613) is 0.7239 tCO₂e/MWh which is higher than 0.5878 tCO₂e/MWh of emission factor for the project. Therefore, the emission factor of the project is conservative. The same has been provided in the updated JPM.				
Documentation provided by project participant				
/1/- version 2.0				
VVB assessment				Date: 16/06/2023
The revised Joint-PD-MR is checked, clarification has been added. Refer to section 3.4.7 of this report for detail assessment of the EF. CL 11 is closed.				

Table 3.CAR from this verification

CAR ID	01	Section no.	1.2	Date : 17/04/2023
Description of CAR				
The description of Sectoral Scope 1 is not in line with the VCS definition of Sectoral Scope.				
Project participant response				Date : 30/05/2023
There is a mistake which has been corrected in the updated JPM.				
Documentation provided by project participant				
/1/- version 2.0				
VVB assessment				Date: 16/06/2023
The revised Joint-PD-MR is checked, CTI confirmed that the Sectoral Scope 1 has been updated in line with the VCS definition of Sectoral Scope 1. CAR 01 is closed.				
CAR ID	02	Section no.	1.16.1	Date : 17/04/2023

Description of CAR	
PP has to provide a declaration of no double counting and provide related information in section 1.16.1.	
Project participant response	Date : 30/05/2023
The relevant information has been added in the updated JPM. The declaration has been provided.	
Documentation provided by project participant	
/1/- version 2.0 /25/	
VVB assessment	Date: 16/06/2023
The revised Joint-PD-MR is checked, CTI confirmed that declaration of no double counting has been provided and clarified. The project is not part of any emission trading program which has been confirmed via checking the UNFCCC/42/, GS/44/, VCS/43/ and other GHG schemes' website. The net GHG emission reductions during this monitoring period from the project was not used for compliance with emission trading programs or to meet binding limits on GHG emissions. The project activity has not participated under any other GHG programs which has been confirmed via checking the UNFCCC/42/, GS/44/, VCS/43/ and other GHG schemes' website, it is verified that the implementation of the project is consistent with the description in the Joint-PD-MR and the monitoring system is fully functional to generate Verified Carbon Units (VCUs) without any double counting for this monitoring period from 27-January-2022 to 31-December-2022 which has been confirmed by checking the Declaration of no double counting and not involved in other GHG scheme/25/. CAR 02 is closed.	

CAR ID	03	Section no.	1.17	Date : 17/04/2023
Description of CAR				
The evidence of the project's SD contributions is not provided in Appendix of JPM.				
Project participant response				Date : 30/05/2023
The part monthly meter records have been provided in the updated JPM. The simplified staff list has been provided in the updated JPM				
Documentation provided by project participant				
/1/- version 2.0				
VVB assessment				Date: 16/06/2023
The revised Joint-PD-MR is checked, CTI confirmed that evidence of the project's SD contributions is provided in Appendix of JPM. Refer to section 3.1 of this report for detail assessment of the SD contributions. CAR 03 is closed.				

CAR ID	04	Section no.	2.1	Date : 17/04/2023
Description of CAR				
1. PP stated that "Though the land is sugar plantation prior to construction, no employment on plantations is reported among major income sources", the related evidence is missing. 2. Why change the sugar plantation to project activity does not have negative impacts on local socio-economy is not specified clearly.				
Project participant response				Date : 30/05/2023
1. The related evidence has been provided. 2. Overwhelming majority of the project area households live mainly on farming. A small number of households also run small businesses. A considerable number of people earn their living as workers in garment industry and as construction workers. Therefore, the project replaces sugar plantation with little impact on the income of local residents. Besides, Salah Essa, a Cambodian individual who owns 6% of the equity of project owner, purchased the project land from previous landowners and then leased it to the project owner for 20 years to operate this solar project. All previous landowners had received total land sold fees and there is no economic dispute. Thus, there are no negative impacts on local socio-economy. The same has been added in the updated JPM.				
Documentation provided by project participant				

/1/- version 2.0	
VVB assessment	Date: 16/06/2023
1. The related evidence has been provided and related description has been added. 2. The explanation has been added accordingly. Refer to section 3.3.1 of this report for detail assessment. CAR 04 is closed.	

CAR ID	05	Section no.	2.2	Date : 17/04/2023
Description of CAR				
PP stated that “the project owner has prepared a stakeholder analysis and communication strategy”, the related evidence is missing.				
Project participant response				Date : 30/05/2023
The relevant evidence has been provided.				
Documentation provided by project participant				

/1/- version 2.0	
/21/	
VVB assessment	Date: 16/06/2023
The revised Joint-PD-MR is checked, CTI confirmed that the evidence of has been added. During the implementation stage of this monitoring period, via checking the Monthly Operation&Maintenance Reports/21/ which include the documented community issues through conducted physical outreach to the neighbouring communities and interview with the local officer/12/ and local residents/13/ during site visit, it is confirmed that the PP has conducted follow-up interviews and ongoing communication with local stakeholders to collect any comments or inputs to the project implementation. CAR 05 is closed.	

CAR ID	06	Section no.	3.1	Date : 17/04/2023
Description of CAR				
The Investment analysis” (version 11.0) is not the latest version. Revision is requested in the whole JPM.				
Project participant response				Date : 30/05/2023
The Investment analysis” (version 11.0) has been replaced by the latest version in the updated JPM.				
Documentation provided by project participant				

/1/- version 2.0	
VVB assessment	Date: 16/06/2023
The revised Joint-PD-MR is checked, CTI confirmed that the version has been updated to 12.0 which is the latest version by checking the UNFCCC website. CAR 06 is closed.	

CAR ID	07	Section no.	3.5	Date : 17/04/2023
Description of CAR				
For sensitivity analysis, 1. The actual investment value of the project is missing and related evidence is missing. 2. How to calculate the annual electricity delivered to the grid based on the installed capacity and operation hours, annual degradation rate and other parameters is not specified. 3. The data source of annual degradation rate need to be specified.				
Project participant response				Date : 30/05/2023
For sensitivity analysis, 1. The actual investment value is 24,595,722 USD and the related evidence has been provided. 2. The electricity delivered to the grid in the first year is calculated as 39,359 MWh based on the installed capacity of 26 MWp and annual photovoltaic electricity yield of 1,513.8 kWh/KWp. By checking the Global Solar Atlas website, the annual photovoltaic electricity yield of the project is reasonable. As per solar PV module specifications provided by manufacturer Q-Cells, the annual				

degradation rate is 0.5%. From the second year onwards, annual electricity delivered to the grid during the whole lifetime is calculated as the electricity delivered to the grid last year minus the electricity reduced due to the degradation of solar PV module, ie $39,162 \text{ MWh} = 39,359 * (1 - 0.5\%)$. The same has been added in the updated JPM. 3. The annual degradation rate is 0.5% which is from the solar PV module specifications provided by manufacturer Q-Cells.	
Documentation provided by project participant	
/1/- version 2.0	
VVB assessment	Date: 16/06/2023
The revised Joint-PD-MR is checked, CTI confirmed that the demonstration has been updated accordingly. Refer to section 3.4.5 of this report for detail assessment. CAR 07 is closed.	

CAR ID	08	Section no.	3.5	Date : 17/04/2023
Description of CAR				
For common practice 1. Sub-Step 4a-2: identify similar projects which fulfil all of the following conditions, PP did not list all the conditions and justify each condition respectively for similar projects' identification. 2. In this section, starting date of the proposed project is stated as 13-Jan-2020 which is not same to other part, revision is requested.				
Project participant response				Date : 30/05/2023
For common practice 1. Each condition has been listed in the section 3.5 and the relevant analysis also has been added in the updated JPM. 2. As the date of the project began generating GHG emission reductions is 27-Jan-2022, the starting date of the proposed project is 27-Jan-2022. The start date has been revised in the updated JPM.				
Documentation provided by project participant				
/1/- version 2.0				
VVB assessment				Date: 16/06/2023
1. The revised Joint-PD-MR is checked, CTI confirmed that PP has listed all the conditions and justify each condition respectively for similar projects' identification. 2. The revised Joint-PD-MR is checked, CTI confirmed that start date has been updated accordingly. CAR 08 is closed.				

CAR ID	09	Section no.	5.1	Date : 17/04/2023
Description of CAR				
For the value listed for annual electricity supplied to the grid $EG_{facility,y}$, how the value is calculated based on the installed capacity and operation hours, annual degradation rate and other parameters is not specified.				
Project participant response				Date : 30/05/2023
The electricity delivered to the grid in the first year is calculated as 39,359 MWh based on the installed capacity of 26 MWp and annual photovoltaic electricity yield of 1,513.8 kWh/KWp. The average operation hours are 1,877h. By checking the Global Solar Atlas website, the annual photovoltaic electricity yield of the project is reasonable. As per solar PV module specifications provided by manufacturer Q-Cells, the annual degradation rate is 0.5%. From the second year onwards, annual electricity delivered to the grid during the whole lifetime is calculated as the electricity delivered to the grid last year minus the electricity reduced due to the degradation of solar PV module. The same has been added in the updated JPM.				
Documentation provided by project participant				
/1/- version 2.0				
/48/				
/14/				

VVB assessment	Date: 16/06/2023
The revised Joint-PD-MR is checked, CTI confirmed that the explanation has been added. Annual electricity delivered to the grid in the first year is calculated as 39,359 MWh based on the installed capacity of 26 MWp and annual photovoltaic electricity yield of 1,513.8 kWh/KWp which has been verified as reasonable by checking the photovoltaic electricity yield data from Global Solar Atlas website/48/. Besides, by checking the technical agreements of solar cell modules provided by manufacturer/14/, it is confirmed that the annual degradation rate of solar cell modules is 0.5%, the annual electricity delivered to the grid during the whole lifetime is calculated as the electricity delivered to the grid last year minus the electricity reduced due to the degradation of solar PV module. CAR 09 is closed.	

CAR ID	10	Section no.	5.4	Date : 17/04/2023
Description of CAR				
The formular for calculation of Net GHG Emission Reductions is not in line with the applied methodology.				
Project participant response				Date : 30/05/2023
The mistake has been corrected in the updated JPM.				
Documentation provided by project participant				
/1/- version 2.0				
VVB assessment				Date: 16/06/2023
The revised Joint-PD-MR is checked, CTI confirmed that LE _y has been deleted accordingly which is verified in line with the formular for calculation of Net GHG Emission Reductions				
CAR 10 is closed.				

CAR ID	11	Section no.	6.3	Date : 17/04/2023
Description of CAR				
For the monitoring plan, 1. The monitoring system diagram missing. 2. How the electricity generated from the solar PV cell modules to finally supplied to the EDC grid is not specified in detail. 3. The electricity meter number is missing and each meter's function is not provided. 4. The national industry standard for meter installation is not provided. 5. The national industry standard for meter calibration is not provided.				
Project participant response				Date : 30/05/2023
For the monitoring plan, 1. The monitoring system diagram has been added in the updated JPM. 2. The detailed information has been added in the updated JPM. 3. The electricity meter number and each meter's function have been added in the updated JPM. 4. The national industry standard for meter installation has been provided in the updated JPM. 5. The national industry standard for meter calibration has been provided in the updated JPM.				
Documentation provided by project participant				
/1/- version 2.0				
VVB assessment				Date: 16/06/2023
1. The revised Joint-PD-MR is checked, CTI confirmed that monitoring system diagram is added. 2. The revised Joint-PD-MR is checked, CTI confirmed that related information of electricity generated from the solar PV cell modules to finally supplied to the EDC grid has been added. 3. The revised Joint-PD-MR is checked, CTI confirmed that electricity meter number is added and their function has been elaborated. 4. The revised Joint-PD-MR is checked, CTI confirmed that national industry standard for meter installation has been described. 5. The revised Joint-PD-MR is checked, CTI confirmed that national industry standard for meter calibration has been described. CAR 11 is closed.				

CAR ID	12	Section no.	7.1	Date : 17/04/2023
Description of CAR				
For the monitoring result of EG_{facility,y} , how the monitoring data for this monitoring period is cross-checked and checked results are all missing.				
Project participant response				Date : 30/05/2023
In this first monitoring period, by cross checking, the monthly meter records of both (a) and (b) were in consistent with the data in the sale and purchase receipts. The same has been added in the updated JPM.				
Documentation provided by project participant				
/1/- version 2.0 /22/ /21/				
VVB assessment				Date: 16/06/2023
The revised Joint-PD-MR is checked, CTI confirmed that information has been added. The monitoring data covering this monitoring period have been cross checked with sale and purchase receipts of electricity/22/. Via comparison, the data in the meter reading records in Monthly Operation&Maintenance Report/21/ are verified as consistent with the data in the sale and purchase receipts of electricity/22/ for both (a) and (b). Hence, VVB confirmed that the meter reading records served as the main data source for emission reductions calculation for this monitoring period is reasonable and correct. CAR 12 is closed.				

Table 4.FAR from this verification

FAR ID	N/A	Section No.	-	Date: -
Description of FAR				
Project participant response				Date: -
Documentation provided by project participant				
VVB assessment				Date: